

SolarEye Security System

Alex Allahar
Victor Igbokwe
Jackson Kitchel
Zayan Prasla

CONCEPT OF OPERATIONS

REVISION – Final Report
Apr 28, 2025

CONCEPT OF OPERATIONS FOR SolarEye Security System

TEAM <65>

APPROVED BY:

Alex Allahar Date

Prof. Kalafatis Date

T/A Date

Change Record

Rev	Date	Originator	Approvals	Description
.				
-	09/11/2024	SolarEye Security System		Draft Release
1	04/28/2025	SolarEye Security System		Final Report

Table of Contents

List of Figures	5
1. Executive Summary	6
2. Introduction	
2.1. Background.....	6
2.2. Overview.....	7
2.3. Referenced Documents and Standards.....	7
3. Operating Concept	
3.1. Scope.....	8
3.2. Operational Description and Constraints.....	8
3.3. System Description.....	9
3.4. Modes of Operations.....	10
3.5. Users.....	10
3.6. Support.....	10
4. Scenario(s)	
4.1. After Hours Vandalism/Trespassing.....	11
4.2. Night-time Accessibility for Certain Parks.....	11
4.3. Kidnapping from a Park.....	11
5. Analysis	
5.1. Summary of Proposed Improvements.....	11
5.2. Disadvantages and Limitations.....	12
5.3. Alternatives.....	12
5.4. Impact.....	12

List of Figures

Figure 1: SolarEye Security System Flow Chart.....	7
Figure 2: Shirewood Park.....	8

1. Executive Summary

Maintaining a safe play area in neighborhoods, parks, and playgrounds is essential for children's well-being and safety. Contrary to using a security officer or relying on bystanders to keep these areas safe, the SolarEye Security System provides an automated surveillance system that can monitor and report potential threats to security. The system will provide lighting and outlets whenever anyone is present in the park, and have the ability to send camera footage to a mobile device via an app. This SolarEye Security System app will provide a user interface including alerts and manual controls of the system. Additionally, this system will be solar powered in order to be self-sustained and off the grid. The system will feature various modes of operation to decrease battery usage during slow periods or after hours. Overall, the SolarEye Security System will provide a more inclusive and safe environment for local neighborhood play areas.

2. Introduction

This document is an introduction to the SolarEye Security System, a system that features an advanced security system designed for playground surveillance. The SolarEye Security System app provides real-time monitoring and alerts users of any detected security threats. With its eco-friendly, automated technology, the system reduces the need for human security patrols, ensuring constant surveillance once the system is in operation.

2.1. Background

Security systems today consist of cameras that are turned on and running continuously. The data from these cameras are stored on-site or are being streamed and monitored in real time by a designated personnel. In order to save money, power, memory storage and manpower, most businesses opt for cheaper systems. Society continues to take steps towards automating remedial tasks.

The SolarEye Security System aims to provide reliable security and surveillance to local neighborhood play areas such as parks and schools. The system will be able to actively monitor the areas of interest with the use of LED lights and a camera. When motion is detected at night during unauthorized hours the LED lights will draw more power from the battery and shine brighter to act as a stronger source of light for people to deter wrong doers by alerting them to the camera that is watching them. Footage from the camera will be accessible through the android application. The system will also power various charging outlets for park goers to be able to charge their devices.

2.2. Overview

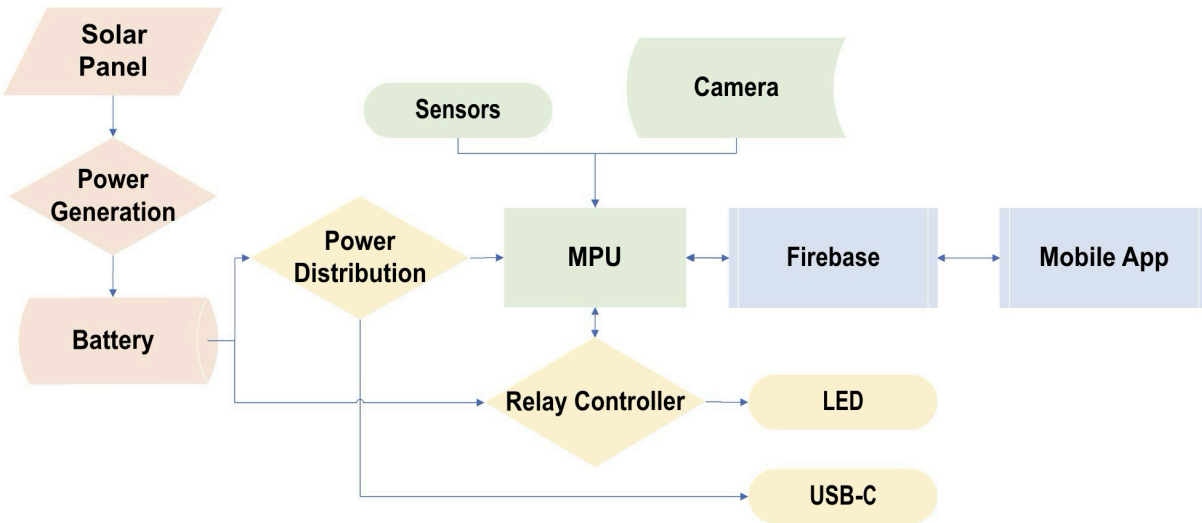


Figure 1: SolarEye Security System Flow Chart

The SolarEye Security System is an advanced, eco-friendly solution for playground security that ensures both active and passive surveillance in local neighborhood play areas. By utilizing automated camera detection and image processing, the system monitors for abnormalities and alerts administrators and users via the SolarEye Security System app, providing event history, and app notifications. The SolarEye Security System features a network to collect image data and notify users, while responsive LED lights and charging outlets enhance user interaction and security on-site. The LED lights adjust brightness when motion is detected at night, serving as a deterrent to trespassers and illuminating the playground for park goers. The system will also power device charging stations to ensure park goers such as parents/guardians have fully charged devices in the case of an emergency. Designed to be self-sustainable, the SolarEye Security System eliminates the need for human surveillance, offering reliable surveillance to protect local play areas in parks and schools.

2.3. Referenced Documents and Standards

- <http://www.solarabcs.org/>
- Solperk 20W solar panel and solar battery charger. <https://solperk.shop/>
- <https://docs.rs-online.com/b1fb/0900766b816fa153.pdf>
- IEEE Wi-Fi communication standards: IEEE 802.11

3. Operating Concept

3.1. Scope

The SolarEye Security System enables self-sufficient automated security of local play areas. Through active surveillance, the area of interest will be monitored for potential security threats, provide recent video footage, and notify park goers with alerts. While motion is detected the camera will be operational. During this time the camera will send near real-time video footage to the microprocessor via a network. Locally, the video data will be analyzed using a basic threat detection model. The microprocessor will relay the security threat detected, and notify users of the SolarEye Security System app. The system will provide a responsive LED to provide active deterrence, depending on the threat detected. The SolarEye Security System's self-sustainable and automated nature will be a versatile solution to increase security and surveillance in local neighborhood playgrounds and parks.



Figure 2: Shirewood Park

3.2. Operational Description and Constraints

The SolarEye Security System is designed to be a self-contained autonomous security system for local, neighborhood playgrounds. The use of solar power will allow this system to function independently of external power systems. The system will implement image processing to detect possible threats, and alert users. A LED will be used as a form of deterrence, based on the type of threat detected. The potential constraints from this operational description are as follows:

- The system needs to be in an environment with sufficient sunlight every 16 hours to continuously power to load devices.

- The playground must be smaller than 300 square feet, for each camera unit to get full coverage.
- If the playground has a canopy it should be at least 12 feet above the base of the playground, to allow for the camera to have full visibility of the playground.
- The budget of 400\$ restricts the quality of materials and parts used to construct the system.
- The system will be designed to function in normal Texas weather conditions, the system's functional in other conditions and climates is not guaranteed.

3.3. System Description

Solar Power Generation: This subsystem provides power to the load devices. The load devices in the system are the camera, outlets, LED, microprocessor, and sensors. The solar panel will charge the battery during the day, which will allow the system to operate at night after authorized playground hours. This subsystem aims to provide an eco-friendly and self-sustained power system to allow this system to not rely on the local play area power infrastructure.

Power Control Subsystem: This subsystem will manage power to load devices to improve system operational longevity and performance. Through the use of an inverter and various power conversion circuits, this subsystem will ensure loads are powered throughout the system. Through the use of switches and sensors, outlets will be automatically turned on or off based on the mode of operation.

Camera Module: This subsystem will manage power of the camera and the transfer of camera video footage. The camera will be triggered by a PIR motion sensor. Once triggered the camera will go from a deep-sleep low voltage mode to an active surveillance mode. A video feed of images will be captured, stored local on an SD card. Near-real time video footage on the SD card will be transferred to the microprocessor via the network for image processing.

Microprocessor and Sensors: This subsystem is responsible for analyzing the sensor data and camera data. Through the use of a microprocessor near real-time video footage can be analyzed for threats. A lightweight classification model, will process the frames of the video data locally on the microprocessors. The model will look for the following security threats: Vandalism, Loitering, and Abnormal or Unsafe Movement. If a threat is detected, an app alert will be sent out and the LED will increase in brightness. The microprocessor will also receive sensor data about the camera, LED, and overall electrical metrics of the system. The sensors will be broken into three modules: Weather Sensor Module, Electrical Sensor Module, and Security Detection Sensor Module. These sensors will determine modes of operation.

Network: This subsystem will be used as the main method of communication between the camera, microprocessor and the app. This network will have WIFI connectivity, to allow the SolarEye Security System App administrator to monitor remotely, and view

recent video footage if a threat has been detected. The network will be encrypted to secure sensitivity video footage of the area of interest.

SolarEye Security System App: This subsystem will be the main interface for users of the playground. Users of the app will be provided near real-time notifications of the park regarding safety. The application will also have administrator access, to allow the owner or operator of the playground to send manual notifications to users, remotely live-stream the play area, and manual control of non-camera load devices.

3.4. Modes of Operations

The SolarEye Security System will have three modes of operation.

The first main mode of operation is the daytime surveillance mode. This mode of operation will have all loads to be at full power. In this mode, an active live feed of the playground will be accessible by the administrator of the SolarEye Application.

The second mode of operation is the nighttime surveillance mode. This mode of operation will only have the LED and camera load active. The camera load will be in a low-voltage sleep state that will be activated by a PIR sensor. Event history will be able to be accessed by the app admin when the camera is activated.

The third mode of operation is the low-power mode. This mode of operation will be activated by either the weather sensors or systems electrical sensors. In this mode, all the load devices will be off except for the LED. When this mode of operation is enabled the administrator will be notified to check the system or unsafe weather has been detected at the playground. The users will also be notified that either the security system is down, or unsafe weather has been detected. In both cases the user will be informed to not go to the park until either events are resolved.

3.5. Users

This system is designed to assist in the security of local play areas found in parks, playgrounds and schools. These areas may not have security guards or are not in proximity to a power system, thus this system can allow these areas to be used safely. Due to the nature of the system, multiple system units could be used in conjunction with an external network to protect larger playgrounds and parks.

3.6. Support

Support for the SolarEye Security System will be provided by the SolarEye Security App. The app will provide an overview of the types of load devices, the meaning of the different modes of operation, and what the LEDs' different brightness mean. Further tutorials on system alerts and a user forum will be provided for added user support.

4. Scenario(s)

4.1. After Hours Vandalism/Trespassing

Implementing a security system in parks and playgrounds can significantly enhance safety by deterring vandalism and preventing illegal activities. The surveillance system, paired with motion detection, not only alerts security teams when needed but also serves as a visible deterrent. This proactive approach ensures the space remains a welcoming and safe environment for families and children while discouraging trespassers from unlawful behavior.

4.2. Night-time Accessibility for Certain Parks

If the security system were to be used for a basketball park, the LED lights can be of assistance to the park people. Communities would benefit from this and more people would be able to access the basketball park with less worry.

4.3. Kidnapping from a Park

If there was an incident involving a child being taken, then the security system will be able to capture the events and be able to show the video of the events. This would better help authorities in finding the suspect and give a jumpstart to the investigation.

5. Analysis

5.1. Summary of Proposed Improvements

- The system will run entirely on solar energy, making it self-sufficient.
- Motion sensors will cause the camera to turn on, stopping the system from continuously recording, using up battery life, and wasting energy on the video processor.
- The system will operate according to a preset schedule and won't need user intervention for maintenance.
- The system will have three different modes, low, normal, and weather. Each mode will maintain the system energy as a predetermined schedule.
- The system will capture all videos and pictures on a SD card and will notify the app.
- The system will notify the app of battery life.
- System will also notify the app of suspicious activity.
- Deterrence by LED light, such as the light will shine brighter than normal for suspicious activity.

5.2. Disadvantages and Limitations

- It is ideal to position the camera unit high to reduce the risk of weather damage or vandalism.

- The system can lose power if the weather tends to be bad for multiple days on end.
- The camera range is limited due to budget constraints and power consumption.
- The camera system is in a fixed position which allows for blindspots.

5.3. Alternatives

- No security system but signs as deterrents to keep playgrounds safe.
- Night time security guards to keep watch of any illegal activities.
- Connection between the camera and the app could be facilitated through a WiFi system with audio and light deterrents.
- The whole security system can be powered through power lines instead of solar panels to ensure steady power to the system.

5.4. Impact

- **Environmental Impact** - The security system is solar powered making it a self-sustaining system using a clean energy source.
- **Societal Impact** - Access to technology is better because having a charging station can help underserved communities. This system can also help with inclusivity within the community by making playgrounds safer for children to visit.
- **Ethical Impact** - Video history of the park can be stolen or tampered with which can lead to issues if an incident were to occur.
- **Mental Impact** - Parents can have a peace of mind knowing their children are at a safe playground with a charging station if they have a device.

SolarEye Security System

Alex Allahar
Victor Igbokwe
Jackson Kitchel
Zayan Prasla

FUNCTIONAL SYSTEM REQUIREMENTS

FUNCTIONAL SYSTEM REQUIREMENTS

FOR
SolarEye Security System

PREPARED BY:

Author Date

APPROVED BY:

Alex Allahar Date

John Lusher, P.E. Date

T/A Date

Change Record

Rev	Date	Originator	Approvals	Description
1	09/26/2024	SolarEye Security System		Draft Release
2	10/04/2024	SolarEye Security System		Diagrams Modified
3	04/28/2025	SolarEye Security System		Final Report

Table of Contents

List of Tables	5
List of Figures	6
1. Introduction	7
1.1. Purpose and Scope.....	7
1.2. Responsibility and Change Authority.....	8
2. Applicable and Reference Documents	8
2.1. Applicable Documents.....	8
2.2. Reference Documents.....	8
2.3. Order of Precedence.....	9
3. Requirements	10
3.1. System Definition.....	10
3.2. Characteristics.....	10
3.2.1. Functional / Performance Requirements.....	10
3.2.2. Physical Characteristics.....	11
3.2.3. Electrical Characteristics.....	12
3.2.4. Environmental Requirements.....	12
3.2.5. Failure Propagation.....	13
4. Support Requirements	13
Appendix A Acronyms and Abbreviations	14

List of Tables

Table1:SubsystemAssignments.....	7
Table2:ReferenceDocuments.....	8
Table 3: Application Documents.....	8

List of Figures

Figure1:SolarEye Security System Concept Image.....	7
Figure2:SolarEyeSecuritySystemBlockDiagram	8
Figure 3: SolarEye Security System Camera Surveillance Flowchart	8

1. Introduction

1.1 Purpose and Scope

The Solar Playground security system, powered by the Solareye Security System, is designed to ensure the safety and well-being of children in parks and playgrounds through automated surveillance technology. Rather than relying on security officers or bystanders, this eco-friendly system actively monitors play areas using cameras, LED lighting, and charging outlets. Motion-triggered LED lights provide enhanced visibility at night, deterring potential wrongdoers by drawing attention to the camera. The app can be used for sending alerts and providing images directly to users' mobile devices, while also offering manual controls and event history. In addition, the system is powered by solar energy, making it self-sustaining and independent from the grid. By eliminating the need for continuous monitoring by personnel or nightly patrols, Solareye Security System provides both active and passive surveillance, reducing power consumption and operational costs. Overall, the Solar Playground system offers a modern, cost-effective solution that enhances security while creating a safer and more inclusive environment for public spaces.

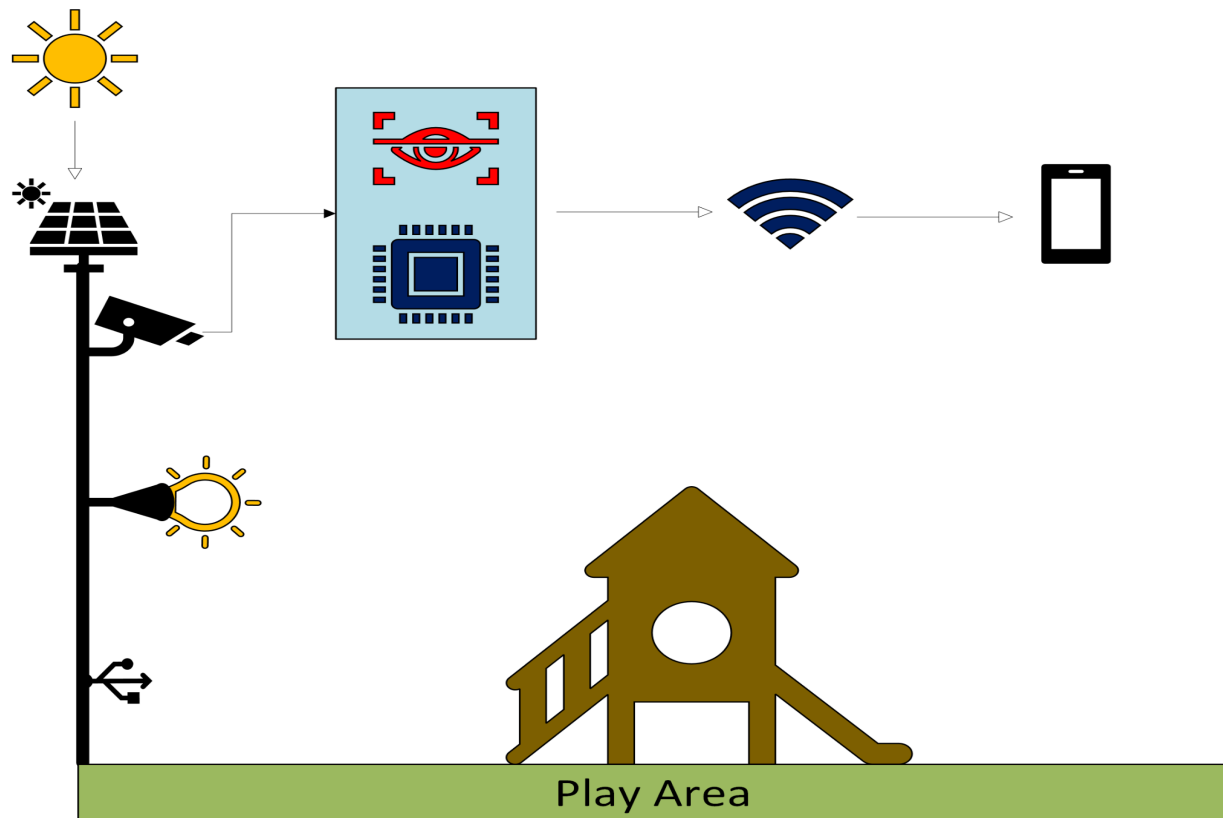


Figure 1: SolarEye Security System Concept Image

1.2 Responsibility and Change Authority

The team leader, Alex Allahar, will have the responsibility of making sure the requirements are met. The requirements can only be changed by the team leader and Dr. Wonhyeok Jang.

Subsystem	Responsibility
Solar Power and Battery	Jackson Kitchel
Power Distribution/Load Management	Victor Igbokwe
Camera, Micronctonroller, and Sensors	Alex Allahar
Network and App	Zayan Prasla

Table 1: Subsystems Assignments

2. Applicable and Reference Documents

2.1 Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein.

Document Number	Revision/Release Date	Document Title
IPC-A-610 J	Revision J - 4/4/2024	Acceptability for Electronic Assemblies
802.11	1/23/1997	Local Area Network(LAN) Technical Standards
1910.305(a)	3/31/2004	Wiring Methods, components and equipment
ANSCI C119.6	6/29/2018	American National Standard for Electric Connectors - Non-Sealed, Multiport Connector Systems Rated 600 Volts or Less for Aluminum and Copper Conductors

2.2 Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

Document Number	Revision/Release Date	Document Title
IEC 61730	09/13/2023	Photovoltaic (PV) module safety qualification
NFPA 855	8/25/2023	Standard for the Installation of Stationary Energy Storage Systems
UL-1642-Fifth Edition	3/13/2012	Standard for Lithium Batteries

2.3 Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings or other documents that are invoked as “applicable” in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICDs that have their relevant documents considered to be incorporated as cited.

3. Requirements

3.1. System Definition

The SolarEye Security System is a cutting-edge, sustainable solution designed to enhance security in local neighborhood playgrounds. It combines both active and passive surveillance methods to monitor play areas effectively. Using automated camera detection and advanced image processing, the system continuously scans for irregularities and immediately alerts administrators and users through the SolarEye app. The app provides access to event history and sends real-time notifications, ensuring timely responses.

The system operates through an integrated network, gathering image data and issuing notifications to users. Its responsive LED lighting system, which brightens when motion is detected at night, serves both as a deterrent to potential intruders and a safety feature for park visitors.

Fully self-sustaining, the SolarEye Security System removes the need for human oversight while offering a reliable security presence to safeguard playgrounds in parks and schools.

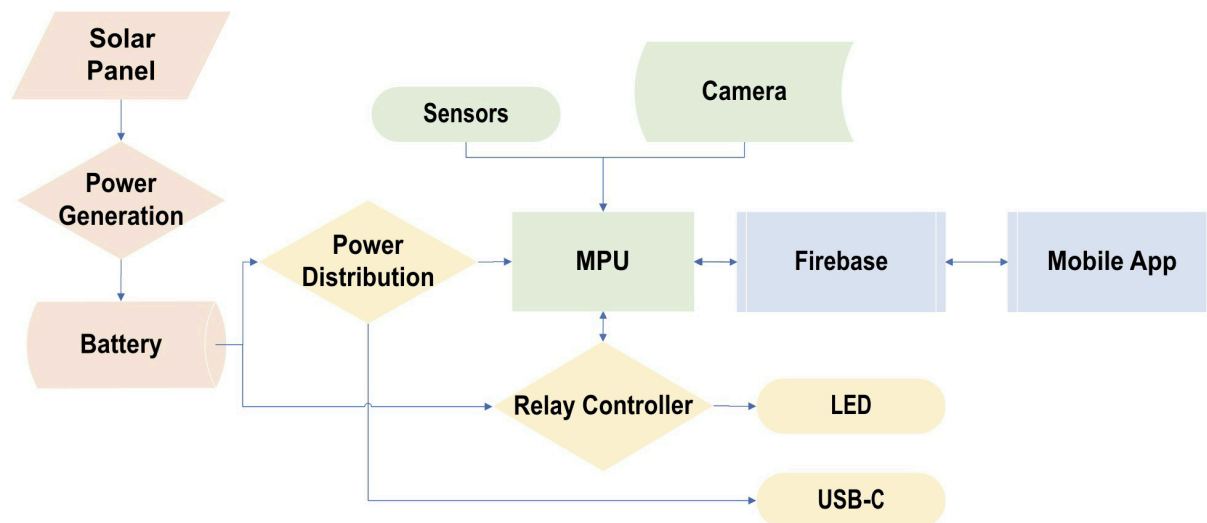


Figure 2: SolarEye Security System Block Diagram

The SolarEye Security System starts with a solar generation subsystem. The solar panel charges the 12V battery. This subsystem is connected directly to the power distribution subsystem. The power distribution subsystem provides the correct voltage and current to the system load devices. The load devices in the system are the IP camera, Raspberry PI, LED, USB-A and USB-C. This subsystem should automatically change power sent to load devices depending on the mode of operation. This subsystem leads into the camera, microcontroller and sensor subsystem. This subsystem is responsible for providing surveillance and capturing information about the playground area of interest. Using a motion sensor to trigger a camera, the video footage will be processed and classified using the Raspberry PI. The Raspberry PI will brighter the LED, and send app alerts via the network. The outputs of the subsystem go to the final subsystem network and App. This subsystem will transmit data from the Raspberry PI and mobile app to all near-real time update about playground security.

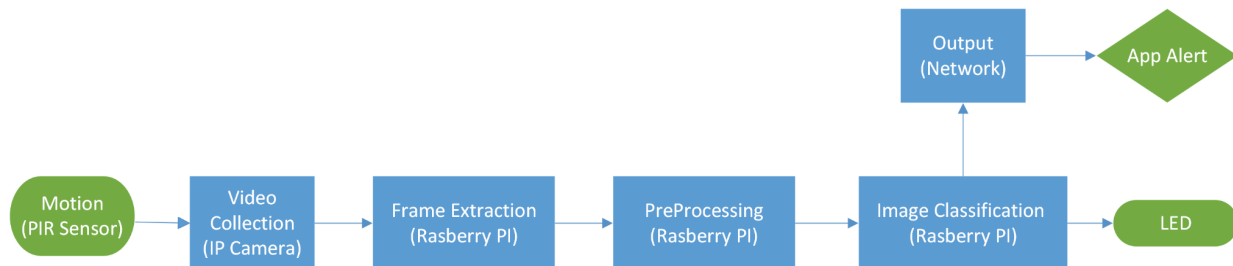


Figure 3: SolarEye Security System Camera Surveillance Flowchart

When motion is detected using the PIR Sensor the camera will record a small video stream of footage. This footage will be sent to the Raspberry PI via a wire. Locally, the frames of video are extracted into images. These images are pre processed and converted into a lower resolution. These new images are then filtered and sent to an image classification inference model, to determine if any abnormal motion or object are detected. Depending on the output of this model, the system's LED will increase in brightness to deter the action detected by the camera. The outputs of the model will also send a JSON output that the network will transmit and push to app alerts to users of the SolarEye Security App.

3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Active Deterrence

The security system will use various LED light levels to act as active deterrence in the playground.

Rationale: The system must be able to provide active security for users at the playground that may not have access to the app.

3.2.1.2. Solar Charging Time (6 hours)

The solar charging system shall provide a full charge to the battery in 6 hours assuming direct sunlight.

3.2.1.3. Time to App Alert

Assuming the Wi-Fi and power source is uninterrupted, it should take no longer than 5 minutes to alert the app.

3.2.1.4. Camera Node FOV

The camera node shall be able to monitor 250-300 foot area with a 120 degree radius.

3.2.1.5. Image Classification Inference Accuracy

The Image Classification inference accuracy should be around 55-65%.

3.2.1.6. Motion Detection accuracy

The motion detection accuracy should be around 65-75%.

3.2.1.7. Input Voltage Level

The input voltage level for the Search and Rescue System shall be +12 VDC.

Rationale: The battery will output +12V VDC into the power distribution circuit which will convert the power for each load.

3.2.1.8. Wiring

The solar security system shall follow the guidelines set forth in the National electrical code regarding electrical wiring.

3.2.1.9. Thermal

The solar security system may be able to function properly in an environment with temperatures ranging 20°F to 110°F

Rationale: The temperature in College station can reach up to 110°F, and the minimum temperature can reach 20°F. Our range ensures the system can properly function in varying climates year round.

3.2.1.10. Connectors

The solar security system shall follow the American National Standard for Electrical Connectors ANSI C119.6-2011.

3.2.1.11. Battery Operating Time

The system should be functional on battery power for 7-8 hours.

3.2.2. Physical Characteristics

3.2.2.1. System Area

The system can provide surveillance over a 250-300 sq.ft provided that the system's camera is installed 6-7 feet in the air and 8-9 feet from the edge of the playground.

3.2.2.2. Installation

Installation manual will be provided to the client.

3.2.2.3. Mounting

Information about mounting will be provided in the Solareye Security System ICD.

3.2.2.4. Water Resistance

The system shall be enclosed in a water resistant case to protect electrical components.

3.2.2.5. Field of View

The field of view of the camera should be 120 degrees.

3.2.3. Electrical Characteristics

3.2.3.1. Outputs

3.2.3.2. Data Output

The SolarEye Security System will have an android application for users to view alerts and should be able to view images.

Rationale: The SolarEye Security System should be readily available to the user through an android application.

3.2.3.3. Raw Video Output

The SolarEye Security System should have an internal SD card to store video footage of daily events.

Rationale: Store longer videos on the SD card.

3.2.4. Environmental Requirements

The SolarEye Security System shall be designed to withstand and operate in the environments and laboratory tests specified in the following section.

3.2.4.1. Pressure (Altitude)

The SolarEye Security System should be able to function properly at altitudes ranging from sea level to 450 feet above sea level.

3.2.4.2. Thermal

The SolarEye Security System should be able to function properly in an environment with temperatures ranging from 20°F to 120°F.

3.2.4.3. Rain

The SolarEye Security System should be functional as the power supply should store electricity use while there is no sunlight during rain.

3.2.4.4. Humidity

The SolarEye Security System may be able to function properly in an environment with relative humidity ranging from 0% to 100%.

3.2.5. Failure Propagation

The Solareye Security System should not allow any faults

4. Support Requirements

4.1.1 Android Phone or Tablet with Internet Connection

In order to access the camera and LED controls as designed, the user must have access to an Android Phone or Tablet with internet connection to view the application.

4.1.2 Shade Free Area

To maintain a steady power supply the system must be kept out in the sun, away from shade for the solar panel to convert sunlight into electricity.

Appendix A: Acronyms and Abbreviations

LED	Light Emitting Diode
PIR	Passive Infrared
FOV	Field of View
USB	Universal Serial Bus

SolarEye Security System

Alex Allahar
Victor Igbokwe
Jackson Kitchel
Zayan Prasla

INTERFACE CONTROL DOCUMENT

INTERFACE CONTROL DOCUMENT FOR SolarEye Security System

PREPARED BY:

Author Date

APPROVED BY:

Alex Allahar Date

John Lusher II, P.E. Date

T/A Date

Change Record

Rev	Date	Originator	Approvals	Description
1	09/24/24	SolarEye Security System		Draft Release
2	10/05/24	SolarEye Security System		LED Hideaway Strobe Lights Removed
1	04/28/2025	SolarEye Security System		Final Report

Table of Contents

List of Tables.....	4
List of Figures.....	5
1. Overview.....	6
2. References and Definitions.....	6
2.1. References.....	6
2.2. Definitions.....	6
3. Physical Interface.....	7
3.1. Weight.....	7
3.2. Dimensions.....	8
3.3. Mounting Locations.....	9
4. Thermal Interface.....	9
5. Electrical Interface.....	9
5.1. Primary Input Power.....	9
5.2. Polarity Reversal.....	10
5.3. Signal Interfaces.....	10
5.4. Video Interfaces.....	10
5.5. User Control Interface.....	10
6. Communications / Device Interface Protocols.....	10
6.1. Wireless Communications (WiFi).....	10
6.2. Host Device.....	11
6.3. Video Interface.....	11
6.4. Device Peripheral Interface.....	11

List of Tables

- Table 1: Solar Generation Weight
- Table 2: Camera and Sensor Unit Weight
- Table 3: Load Devices Weight
- Table 4: Solar Generation Unit Dimensions
- Table 5: Camera and Sensor Dimensions
- Table 6: Load Devices Dimensions
- Table 7: Maximum Voltage and Current Levels
- Table 8: Stand-by Voltage and Current

List of Figures

Figure 1: Electrical Interface Diagram

1. Overview

This document provides the details for the SolarEye Security System. It will list the inputs and outputs of the system and detail how the system manages them. The parts that are intended to be used to design the system are mentioned in this section of the document. Attributes of the parts such as physical, electrical, and thermal, communication and device interface protocols are described.

2. References and Definitions

2.1 References

Refer to section 2.2 of the Functional System Requirements document.

2.2 Definitions

oz	ounces
m	meters
in	inches
mA	Milliamp
mW	Milliwatt
MHz	Megahertz (1,000,000 Hz)
TBD	To Be Determined

3. Physical Interface

3.1 Weight

3.1.1 Solar Generation Weight

Component	Weight	Number of Items	Total Weight
12.8V 50Ah 640Wh LiFePO4 Lithium Battery	~20lbs (TBD)	1	~20lbs (TBD)
Charging Circuit PCB	TBD	1	TBD
75W-100W Solar Panel	TBD	1	TBD

3.1.2 Camera and Sensor Unit Weight

Component	Weight	Number of Items	Total Weight
Raspberry Pi 4 B	0.76 oz	1	0.76 oz
Ansice IP Camera	15.9 oz	1	15.9 oz
DaierTek Battery Voltage Meter	1.76 oz	1	1.76 oz
Raspberry Pi 15W USB-C Power Supply	5.13 oz	1	5.13 oz

3.1.3 Load Devices Weight

Component	Weight	Number of Items	Total Weight
Buck Converter PCB	TBD	1	TBD
USB Charging Ports	TBD	3	TBD

3.2 Dimensions

3.2.1 Dimension of Solar Generation

Component	Length	Width	Height
12.8V 50Ah 640Wh LiFePO4 Lithium Battery	7.68 in	6.42 in	6.61 in
Charging Circuit PCB	TBD	TBD	TBD
75W-100W Solar Panel	TBD	TBD	TBD

3.2.2 Dimension of Load Devices

Component	Length	Width	Height
Buck Converter PCB	TBD	TBD	TBD

3.2.3 Dimension of Camera and Sensors Unit

Component	Length	Width	Height
Raspberry Pi 4 B	3.37 in	2.22 in	0.67 in
Ansice IP Camera	8.54 in	4.41in	3.23 in
DaierTek Battery Voltage Meter	6.57 in	4.29 in	1.06 in
Raspberry Pi 15W USB-C Power Supply	1.5 m	TBD	TBD

3.3 Mounting Locations

The system is meant to be mounted on a pole minimum 6-7 feet above the ground. The camera field of view is 120 which should cover a minimum of 250 sq ft. The system must be clear of any shade/coverings so the solar panel can successfully charge the battery and power the system, but also accessible for the charging station.

4. Thermal Interface

The Raspberry Pi 4 B will use a heat sink that has a cold wall. Since the Raspberry Pi 4 B will be responsible for analyzing video footage, hosting the database, collecting data on the sensors, and transmitting data to the users of the mobile app, a heat sink will be needed to prevent overheating. Due to the need of the security system to provide near real-time updates on the park's security, overheating could lead to a decrease in efficiency of the Raspberry Pi and thus a decrease in security of the playground.

4.1 Battery Cooling

The battery will be enclosed in a well-ventilated housing, ensuring efficient dissipation of excess heat.

5. Electrical Interface

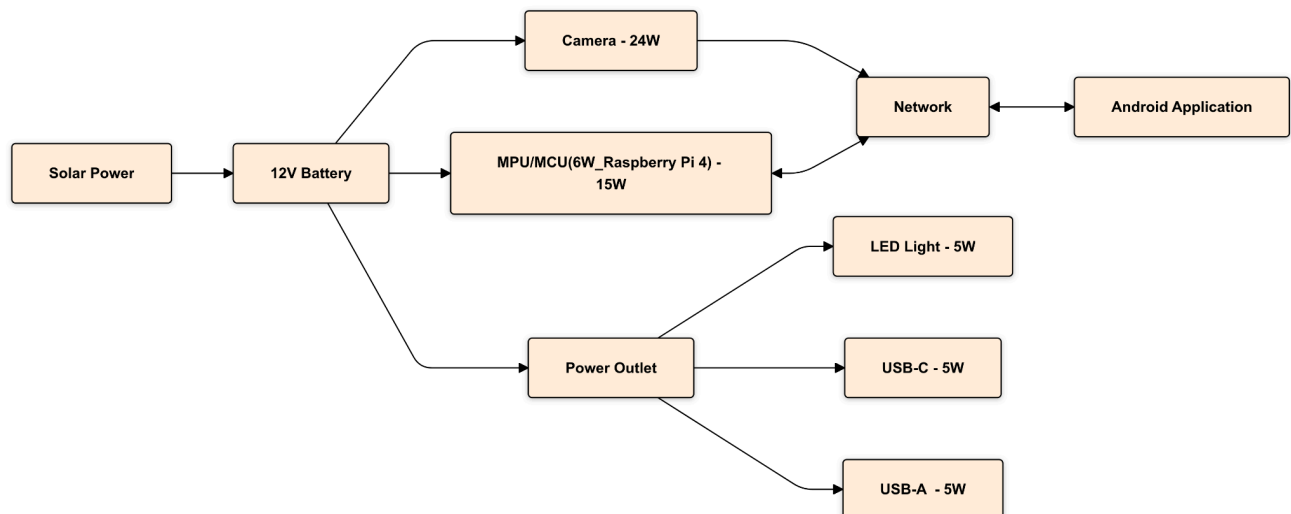


Figure 1: Electrical Interface Diagram

5.1 Primary Input Power

The Primary input power will be a 12 volt Lithium battery, that will be charged by a 90 Watt solar panel.

5.2 Voltage and Current Values

5.2.1 Maximum Values

Component	Voltage(V)	Current(A)	Power(W)
USB Charging	5	1	5

Ports			
LED Hideaway Strobe Lights	TBD	1	TBD
Raspberry Pi 4	TBD	TBD	TBD
Sensors	TBD	TBD	TBD
Camera	12	2	24
Battery	12.8V	≤50A	640Wh
Solar Panel	TBD	TBD	90W

5.3 Signal Interfaces

The camera and the microcontroller are connected via wired connection. The microcontroller has a general network adapter which includes WiFi Network.

5.4 User Control Interface

The user interface is a web application on Android devices only. The application will connect to the main control unit or the main microprocessor for the system. The user interface should present battery charging statistics, predict charge patterns, and predict when the battery life ends. There should be AI suggestions/reminders to help the user check on the system. There will be a secure login to keep the application in safe hands.

6. Communications / Device Interface Protocols

6.1 Wireless Communications (WiFi)

The microcontroller has a built-in WiFi module using IEEE 802.11 g/b/n standards. This connection will be used to send alerts to the user application using serial communication.

6.2 Host Device

The host device will be the Raspberry Pi 4 B, which will host its own server/database.

6.3 Device *Peripheral Interface*

The Camera has a built-in PIR Sensor to detect motion. Once triggered the camera will record and transmit video footage to the Raspberry Pi to begin the security system process.

SolarEye Security System

Alex Allahar

Victor Igbokwe

Jackson Kitchel

Zayan Prasla

EXECUTION PLAN

TASK TITLE	TASK OWNER	START DATE	DUE DATE	Status
Concept of Operations	All	9/8/24	10/15/24	✓
Functional System Requirements	All	9/15/24	9/26/24	✓
Interface Control Document	All	9/15/24	9/26/24	✓
Midterm Presentation	All	9/27/24	10/2/24	✓
Solar Generation Circuit Design	Jackson	10/2/24	10/15/24	✓
Power Circuit Design	Victor	10/2/24	10/15/24	✓
Learn about Machine Learning	Alex	10/2/24	10/15/24	✓
Create Dev Account (PlayStore)	Zayan	10/2/24	10/15/24	✓
Subsystem Introduction Project	All	10/2/24	10/15/24	✓
Project Parts Ordered	All	10/2/24	10/15/24	✓
Learn Kotlin	Zayan	10/2/24	10/15/24	✓
Design Security for App	Zayan	10/10/24	10/20/24	✓
Test Electrical Sensors with MPU	Alex	10/15/24	10/24/24	✓
Status Update Presentation	All	10/15/24	10/25/24	✓
Redesign UI login/create account page of App	Zayan	10/15/24	10/25/24	✓
Design UI main page of App	Zayan	10/15/24	10/25/24	✓
Altium PCB Design	Jackson	10/15/24	10/29/24	✓
Altium PCB Design	Victor	10/15/24	10/29/24	✓
Pi to Server Connection	Alex Zayan	11/1/24	11/10/24	✓
Test Parts w/ Breadboard and Power	Jackson	10/22/24	10/30/24	✓
Test Parts w/ Breadboard and Power	Victor	10/22/24	10/30/24	✓
Connect App to Microprocessor (Test)	Zayan	10/28/24	11/8/24	✓
Server to Firebase Connection	Alex	10/20/24	10/26/24	✓
Order PCB	Jackson	10/29/24	11/3/24	✓
Order PCB	Victor	10/29/24	11/3/24	✓
Test and Publish App on Google PlayStore	Zayan	11/1/24	11/14/24	✓
Camera connected to MPU	Alex	10/27/24	11/8/24	✓
Video Capture to Firebase	Alex	11/1/24	11/8/24	✓
Solder PCB and Test	Jackson	11/3/24	11/18/24	✓
Solder PCB and Test	Victor	11/3/24	11/18/24	✓
Motion Detection Mode	Alex	11/3/24	11/8/24	✓
Live Streaming Mode	Alex	11/10/24	11/15/24	✓
Final Presentation	All	11/4/24	11/20/24	✓
Final Demo	All	11/4/24	11/26/24	✓
Final Report	All	11/4/24	12/5/24	□

Tasks	5-Feb	12-Feb	19-Feb	26-Feb	5-Mar	19-Mar	26-Mar	02-Apr	09-Apr	16-Apr	Date
Status Update 1											29-Jan
Reorder-PCBs											5-Feb
Solder Power Distribution PCB											5-Feb
TPEC Poster Presentation											12-Feb
App Backend Security											12-Feb
MPU Database / App Database Integration											12-Feb
Status Update 2											12-Feb
App and Network Integration											19-Feb
Live Streaming from MPU to App (Alex & Zayan)											26-Feb
Buck Converter/MPU integration											26-Feb
Solder/Test Relay controller PCB											26-Feb
Status Update 3											26-Feb
Relay Controller / MPU Integration (Victor & Alex)											19-Mar
Validate LED / MPU Communication (Alex)											19-Mar
Validate Buck converter w/ MPU livestream (Victor)											3/19/2025
App Frontend UI Update (Zayan)											19-Mar
Charging Statistics to App Integration (Jackson)											2-Apr
MPU Modes of Operation App Control											2-Apr
Test Power Generation PCB (Jackson)											4/2/2025
Integrate Power Generation and Distribution (Jackson & Victor)											4/2/2025
Status Update 4											4/2/2025
Start Housing (All)											4/2/2025
Video Footage App / MPU Validation (Alex)											4/2/2025
Connect cooling (All)											4/9/2025
Validate system solar generation Components (All)											4/9/2025
Status Update 5											4/2/2025
MPU / App Livestream (Zayan & Alex)											4/16/2025
Validate Video Player (Zayan)											4/16/2025
Sensor Communication to MCU (Jackson & Alex)											
Charging Statistics (Zayan)											4/16/2025
Mount PCBs to system housing (All)											4/16/2025
Replace Camera (Alex)											4/9/2025
Final Presentation											4/16/2025
System Communication Range (All)											4/16/2025
Outdoor System Mode of Operation Testing											4/20/2025
System Lifespan Testing											4/23/2025
Final Demo											
											Completed
											In Progress
											Not Started

SolarEye Security System

Alex Allahar

Victor Igbokwe

Jackson Kitchel

Zayan Prasla

VALIDATION PLAN

Test Name	Success Criteria	Methodology	Status	Assigned to
PIR Sensor Range	Range > 5 ft, FOV >= 120 degrees	Run PIR_validation.py and see if the PIR Sensor can detect motion from at least 5 ft., and at least 120 degrees from face of the sensor.	Completed	Alex Allahar
PIR Sensor to Firebase Response Delay	Avg Response Time = $(0.2039+0.0671+0.0609+0.0706+0.0632+0.0616+0.0632+0.0616+0.0625+0.0694) / 10 = 0.0784$ seconds	Using PIR_validation.py to monitor motion on the GPIO Pin assigned to the PIR Sensor, ensure that the sensor data is recieved by firebase, within 5 seconds. Average 10 trials to get accurate metric.	Completed	Alex Allahar
USB Camera to Firebase Delay	Video Duration = 30.5509 seconds. Upload duration = 1.2951 seconds. Total delay = 31.8460 seconds	Using CAM_validation.py to test if the USB Camera can send a 30 second video to Firebase, and determine time between video capture and firebase acknowledgement.	Completed	Alex Allahar
PIR to Detection Automatic Delay	Delay = 0.2082 seconds	Using main.py determine time between PIR sensor detecting motion and USB camera beginning video capture.	Completed	Alex Allahar
Event History	Both MPU and Firebase has up to a week of footage.	Verify that the MPU and Firebase after reboot has the up to 24 of prevent footage.	Completed	Alex Allahar
Video Streaming	The MPU live-stream can be access via HTTP Protocol.	Use UDP to Stream the MPU via HTTP Flask Serv	Completed	Alex Allahar
System Resources	Ensure low-power mode is below operational mode.	Measure Memory and CPU Usage during modes of operation, ensure low-power mode is below operational mode.	Completed	Alex Allahar
Buck Converter Parts Test	All Capacitors and Resistors are providing desired results.	Set up the Converter on a breadboard and test it using the Power supply provided in the lab.	Completed	Victor Igbokwe
PCB Design Test	PCB design is plausible.	Simulation using Sim Software	Completed	Victor Igbokwe
PCB Functionality	PCB Soldered and Functioning	Connect it to the DC Power Supply and Test the components	Completed	Victor Igbokwe
System Power Check	All parts of the system have power and are able to carry out duties.	Test power levels on load devices using a multimeter.	Failed	Victor Igbokwe
Network Connectivity	Able to receive and send signals to and from the app/system.	MPU data is able to be sent and seen on the app.	Completed	Zayan Prasla
Password Policy	User has to put in a password that follows the password policy	Check if password can be created by not following the policy: 6 minimum characters, 1 uppercase and 1 special character	Completed	Zayan Prasla
Store and Fetch Data	Application is able to store and fetch login data.	Test logging into the app with an already created account	Completed	Zayan Prasla
Email Verification	After creating an account, user recieves a verification email.	Check for a verification email being sent after creating an account	Completed	Zayan Prasla
Main Page UI	Successfully display Event History, Charging statistics, and system controls	Check each function on main page and make sure there is no input lag	Completed	Zayan Prasla
Solar Generation Check	A charge rate of 0.45kW.	Test solar charge rate with a multimeter.	Pending	Jackson Kitchel
Operation Time	8 hours of battery life	Monitor the battery after it finishes fulling charging.	Pending	Jackson Kitchel
High Voltage Cut-Off	Charging circuit cuts off when voltage level reaches 15V	Monitor the charging voltage after battery is fully charged	Pending	Jackson Kitchel
Low Voltage Cut-Off	Battery starts charging when low-voltage of 12.4V	Monitor the charging voltage after charge level is low	Pending	Jackson Kitchel
PCB Functionality	PCB Soldered and functioning as required by previous checks	Validate every requirement after PCB is complete	Partial Completion	Jackson Kitchel
PCB Design	PCB Design is plausible.	Simulation Results using software	Completed	Jackson Kitchel

Validation Plan 404			
Test Name	Success Criteria	Methodology	Status
Real Event vs. Video Capture Ending	The system should upload the recorded video to Firebase within 5 seconds of the motion event ending.	A controlled motion event (e.g., waving a hand) was triggered in front of the PIR sensor. The stopwatch was used to measure the time from the end of motion detection to the completion of the video upload to Firebase, based on logged timestamps.	Passed
Livestream Delay	The live video feed should reflect real-time activity with less than 5 seconds of delay.	A stopwatch was used to measure the time between a real-time action (such as waving) and its appearance on the livestream feed via the client interface. This was repeated for consistency.	Partial Completion
Motion Detection Range	The PIR sensor must detect motion at a distance of at least 12 feet from the sensor.	A subject walked slowly toward the sensor from a distance of 15 feet. The point at which the PIR sensor reliably triggered motion detection was recorded. This test was repeated to confirm consistent triggering distance.	Passed
Fans Power Consumption	The total power consumption of all cooling fans must remain under 5 watts .	Voltage and current draw from the fans were measured using a multimeter. Power consumption was calculated using the formula: $P = V \times I$. Measurements were taken during normal operation.	Partial Completion
Pi Power Consumption (Low Power)	The Raspberry Pi should consume less than 20 watts when running in low-power mode.	The Pi was configured with minimal background services, disabled HDMI output, and reduced clock speed. Power draw was measured using a USB inline power meter during idle conditions.	Passed
Pi Power Consumption (Default)	Power consumption should stay below 20 watts during regular operation.	The Pi ran with a standard configuration (GUI, networking, and essential services). Average power draw was logged using a USB power meter over a 10-minute observation window.	Passed
Pi Power Consumption (Livestream)	Total power usage must remain under 20 watts during active video livestreaming.	While running the livestreaming script, power consumption was continuously measured using a USB inline power meter, capturing peak and average values.	Passed
Charging Statistics to Firebase	The system should transmit updated charging current and voltage statistics to Firebase in less than 1 second from measurement.	A microcontroller read voltage and current values from the battery via GPIO-connected sensors. These readings were formatted and pushed to Firebase Realtime Database, with timestamps used to calculate transmission latency.	Completed
Voltage and Current Sensors Communicate successfully to Pi	The battery is successfully being charged by 13 Volts coming out of the charging PCB. Voltage level visible on the Application	Connect the Solar panel to the Charging PCB, which is then connected to the Battery to enable charging. ADC used to obtain voltage readings from sensors, which was then communicated to the Pi and then the application.	Completed
Password Policy	App enforces password rules	Try creating a password with less than 6 characters, no uppercase, no special character, and no number.	Completed
Password Reset	Password reset email is sent	Use "Forgot Password" and check inbox for reset link	Completed
Store and Fetch Data	Application is able to store and fetch login data.	Test logging into the app with an already created account	Completed
Email Verification	User receives a verification email after signup	Create an account and check inbox for verification email	Completed
Login Restriction	Only verified users can access the app	Attempt login before verifying email	Completed
Main Page UI	Successfully display Event History, <i>Charging statistics</i> , and system controls	Check each function on main page and make sure there is no input lag	Completed
Thumbnail Updates	Latest event image is shown on homepage	Trigger new video upload, check if homepage thumbnail updates	Completed
Charging Graph	Displays recent voltage data correctly	Confirm chart updates with last 24 hrs of voltage data	Completed
UI Loads Properly	Main screen elements load without lag	Navigate to home and interact with UI	Completed
Live Streaming	Live stream footage directly from the system camera to the app	Check if live stream is not too behind from actual live camera time	Completed
Modes of Operation	Each mode updates system state in Firebase	Tap each mode and check Firebase document for correct mode value	Completed
Camera Control	Camera state updates on/off through the app	Press camera toggle, verify Firebase update	Completed
Light Control	Lights can be toggled via app and Firebase	Use app buttons to turn lights on/off, confirm state change	Completed
Media Transfer	Successfully transfer pictures and videos with the correct date/time	Check the event history page and make sure picture and video align with date and time	Completed

Network Communication	Video and photo entries show correct timestamps	Confirm data appears from MPU and responds to control actions	Completed ▾
Data Storage	User and sensor data is reliably stored and fetched	Sign in and read sensor values from Firestore	Completed ▾
Realtime Updates	Camera feed shows with low latency	Activate stream, compare delay with real-time movement	Completed ▾
Buck LineTest	Output voltage remains constant with different input voltages	Change input voltage and record output voltages	Completed ▾
Buck Load Test	Output voltage remains constant as the output current changes.	Use Electronic load device	Completed ▾
Relay PCB Load Test	Output voltage remains constant as the output current changes.	Use Electronic load device	Completed ▾
Relay control module Integration	Current flow can be controlled by MPU	Connect GPIO pins on PCB to MPU	Completed ▾
Buck Converter integration with Solar Generation subsystem	Buck converter powered by solar generation system	Connect buck PCB to battery	Completed ▾
Solar Generation Check	A charge power rate of 10W in ideal sunlight.	Test solar charge rate with a multimeter.	Completed ▾
Operation Time	8 hours of battery life	Monitor the battery after it finishes fulling charging.	Completed ▾
High Voltage Cut-Off	Charging circuit cuts off when voltage level reaches 14.8V	Monitor the charging voltage after battery is fully charged	Completed ▾
Low Voltage Cut-Off	Battery starts charging when low-voltage of 12.4V	Monitor the charging voltage after charge level is low	Completed ▾
Voltage Sensors communicate accurately with Raspberry-Pi	Voltage from solar panel and battery match voltage in MCU console	Measure Voltage with mulimeter and compare to MCU console	Completed ▾
PCB Functionality	PCB Soldered and functioning as required by previous checks	Validate every requirement after PCB is complete	Completed ▾
PCB Design	PCB Design is plausible.	Simulation Results using software	Completed ▾
System Housing	The System is able to function and hold all the needed subsystems.	Test All Modes of Operation with the completed housing for the system installed.	Completed ▾
Complete system Test	The complete system with all working subsystems is able to function as expected	Test All Modes of Operation with the complete system	Completed ▾

SolarEye Security System

Alex Allahar

Victor Igbokwe

Jackson Kitchel

Zayan Prasla

SUBSYSTEM REPORTS

SUBSYSTEMS REPORTS FOR SolarEye Security System

PREPARED BY:

Author Date

APPROVED BY:

Alex Allahar Date

John Lusher II, P.E. Date

T/A Date

Change Record

Rev	Date	Originator	Approvals	Description
.				
-	09/24/24	SolarEye Security System		Draft Release
1	04/28/2025	SolarEye Security System		Final Report

Table of Contents

List of Figures.....	51
List of Tables.....	52
1. Introduction.....	53
2. Solar Generation Subsystem Report.....	54
2.1. Subsystem Introduction.....	54
2.2. Subsystem Design.....	54
2.3 Subsystem Details.....	55
2.3.1.Maximum Power Point Tracking.....	55
2.3.2.Current Sensors.....	56
2.4. Subsystem Validation.....	57
2.5.Subsystem Conclusion.....	61
3. Power Subsystem Report.....	62
3.1. Subsystem Introduction.....	62
3.2. Subsystem Details.....	63
3.2.1 Buck Converter PCB.....	64
3.2.2 Relay Control Module PCB.....	66
3.3. Subsystem Validation.....	67
3.4. Subsystem Conclusion.....	71
4. Edge Processing Unit Subsystem Report.....	72
4.1. Subsystem Introduction.....	72
4.2. Subsystem Details.....	72
4.2.1 PIR Sensor.....	72
4.2.2 ADC.....	73
4.2.3 Camera.....	74
4.2.4 Cooling System.....	75
4.2.5 LED Deterrence.....	75
4.2.6 Low Power Mode.....	76
4.2.7 Motion Detection Mode.....	76
4.2.8 Live Streaming.....	79
4.2.9 Firebase Database.....	80
4.3. Subsystem Conclusion.....	81
5. App Subsystem Report.....	82
5.1. Android Application and Database Introduction.....	82
5.2. Android Application Details.....	82
5.2.1. User Login Security.....	82
5.2.2. Home/Settings Page.....	83
5.2.3 Event History.....	83

5.3. Android Application Validations.....	84
5.4. Android Application Conclusion.....	88

List of Figures

Figure 1: Solar Charging PCB Layout.....	54
Figure 2: Soldered Charging Circuit.....	55
Figure 3: Solar Power Charging Circuit Schematic.....	56
Figure 4: Solar Power Charging Circuit Efficiency vs Output Voltage.....	56
Figure 5: Current Sensor Schematic.....	57
Figure 6: Simulated Solar Inputs vs Charging Voltage.....	59
Figure 7: Multimeter vs MCU Readings Line Chart.....	60
Figure 8: Buck Converter Altium Circuit Schematic.....	63
Figure 9: Buck Converter.....	63
Figure 10: Relay Control Module Altium Circuit Schematic.....	64
Figure 11: Figure 11: Relay Control Module PCB.....	64
Figure 12: Buck Voltage regulation.....	65
Figure 13: Buck Load Regulation.....	66
Figure 14: Power Consumption Validation.....	67
Figure 15: Relay Control Module Load regulation.....	68
Figure 16: Relay Control Module Technical block diagram.....	69
Figure 17: MOSFET Switching Times as a Function of Load Current.....	70
Figure 18: Cooling System.....	74
Figure 19: LED Deterrence Active.....	76
Figure 20: Motion Detection Mode Pipeline Terminal.....	77
Figure 21: Livestream UI Layout.....	79
Figure 22: Firestore Motion Detection Data Structure.....	80
Figure 23: Login page showing email verification is needed, Create an Account page, email verification message. (APP VERSION ECEN 403).....	84
Figure 24: Login page with Password Reset, Create an Account page, email verification message. (APP VERSION ECEN 404).....	84
Figure 25: Application home page, Event History page, Settings Page (APP VERSION ECEN 403).....	85
Figure 26: Application Home page with Charging Stats/Modes of Operation/System Controls, Event History page with Live Stream option, Settings Page with FAQ (APP VERSION ECEN 404).....	85
Figure 27: Modes of Operation for power consumption. (APP VERSION ECEN 404).....	86
Figure 28: System Controls, Camera & LED are only operational not functional (APP VERSION ECEN 403).....	86
Figure 29: System Controls, Camera & LED are only operational not functional (APP VERSION ECEN 404).....	87

List of Tables

Table 1: Charging Circuit MPPT Testing.....	58
Table 2: Voltage and Current Sensors MCU vs Multimeter Values.....	59
Table 3: Voltage Drop Under Load.....	61
Table 4: RCM Voltage Drop Under Load.....	63
Table 5: Motion Detection Database Acknowledgement Delay.....	67
Table 6: Video Upload Latency.....	69
Table 7: Real Event vs. Video Capture Delay.....	74
Table 8: Cooling System Current Measurements.....	74
Table 9: Sensor to Camera Delay.....	78
Table 10: Live Stream Delay.....	79

1. Introduction

The SolarEye Security System autonomously monitors playground and park security by collecting data from sensor and camera modules, centralizing it, and uploading it to an online platform accessible via a mobile app. The system is divided into the following subsystems: Solar Generation, Power, Microprocessor, and data SolarEye Security App. These subsystems were specified in the Conops, FSR, and ICD. These subsystems are expanded upon below.

2. Solar Generation Subsystem Report- Jackson Kitchel

2.1 Subsystem Introduction

The Solar power generation subsystem serves as the central power source within the project. The subsystem is responsible for the solar panel, charging circuit PCB and supplying power to the 12-volt battery. The 100W solar panel is connected directly to the charging circuit PCB which regulates the power and supplies the 12-volt battery with a safe and efficient supply voltage. The power from the solar panel is used efficiently by using a Maximum Power Point Tracking (MPPT) IC chip. Additionally, the voltage and current supply from the solar panel as well as the battery are monitored by the charging circuit using current sensors and voltage sensors in combination with an ADC. These data points are then output by the PCB to be read and processed by the Raspberry Pi 4B.

2.2 Subsystem Design

The charging circuit PCB was designed initially using LTspice in which the initial iteration and testing were done. After thorough validation of the design, hardware and specifications were selected and the circuit was then designed in Altium. After careful redesign and consolidation, the final PCB layout can be seen below. Following the final iteration of the Altium PCB design, the parts and PCB were ordered. Lastly, the PCB had to be soldered and tested before being integrated into the system with the other subsystems.

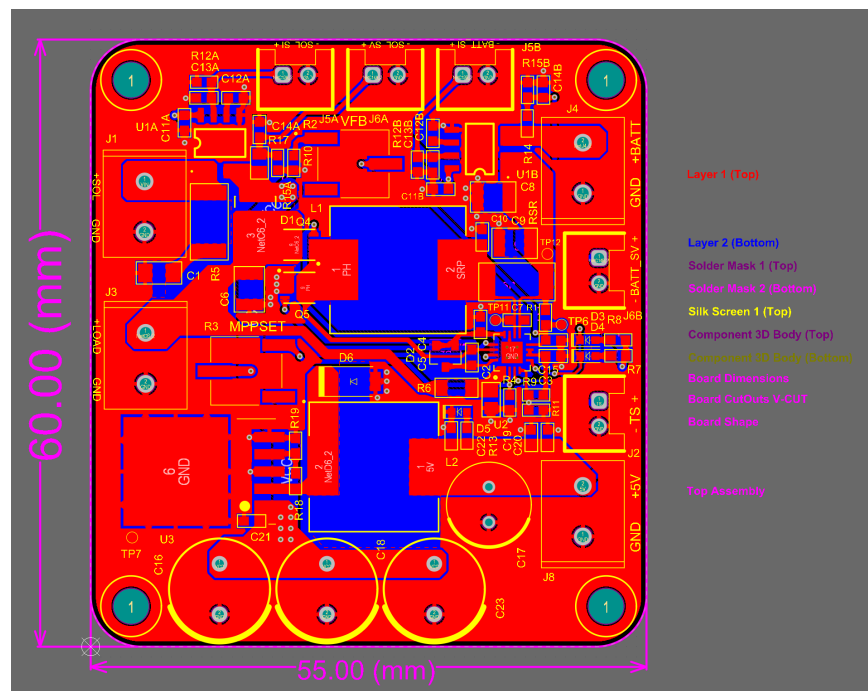


Figure 1: Solar Charging PCB Layout

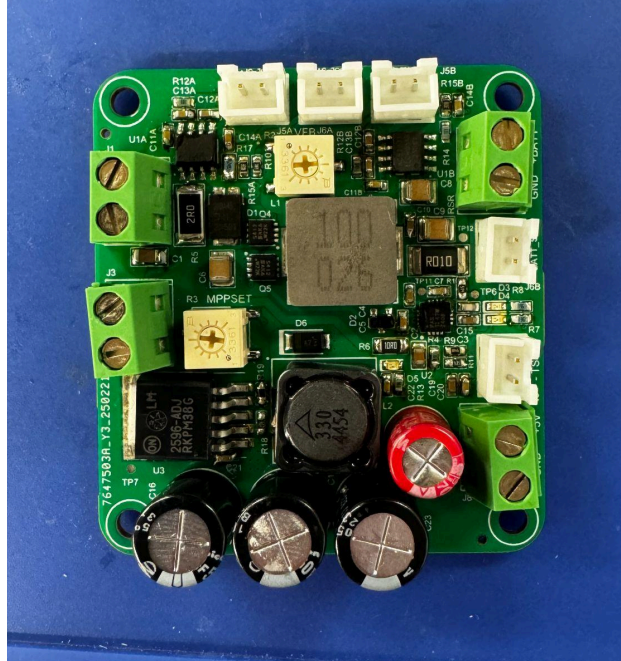


Figure 2: Soldered Charging Circuit

2.3 Subsystem Details

The charging circuit uses several components and IC chips to properly regulate the power supply. The first is the BQ24650, which serves as the Maximum Power Point Tracking (MPPT) aspect of the circuit. This ensures the circuit is able to provide the battery with maximum power, depending on the input voltage of the solar panel. This will change throughout the day contingent upon the sunlight and weather conditions. Furthermore, the circuit incorporates 2 ACS723 current sensor IC chips to monitor the current and voltage at the input and output of the circuit. Details of these components and schematic setup are as follows:

2.3.1 Maximum Power Point Tracking

The BQ24650 serves many purposes in the circuit with the main function being the maximum power point tracking. This allows the battery to be charged at the most efficient rate, while minimizing power loss. To achieve this, the IC chip regulates the charge voltage of the battery based upon the input voltage of the solar panel. Furthermore, the BQ24650 facilitates the high/low cutoff for the charging circuit. In this case, the charging circuit will cut off voltage to the battery when the battery exceeds 15V (cut off voltage). It will also ensure that the circuit begins charging the battery when the battery falls to 12.4V (cut on voltage). These measures will ensure that the battery is never overcharged, while also maintaining charge within the battery. Furthermore, the BQ24650 incorporates temperature protection via a thermistor. This ensures that the battery will never overheat while being charged in order to prevent damages and

hazards. The schematic setup of the BQ24650 can be seen below along with its efficiency curve:

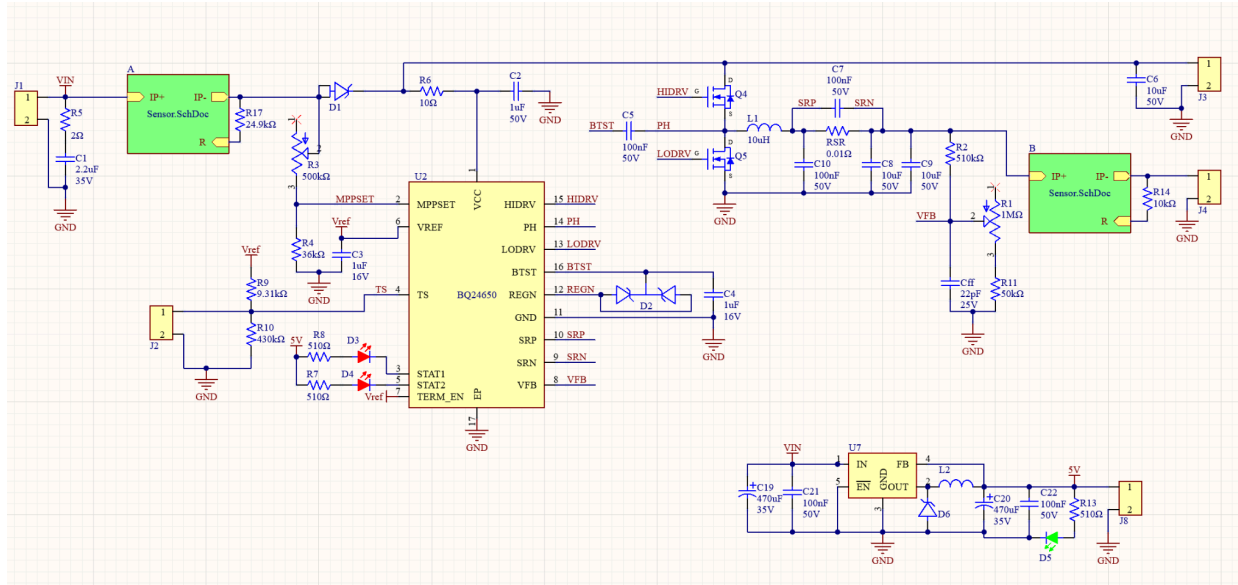


Figure 3: Solar Power Charging Circuit Schematic

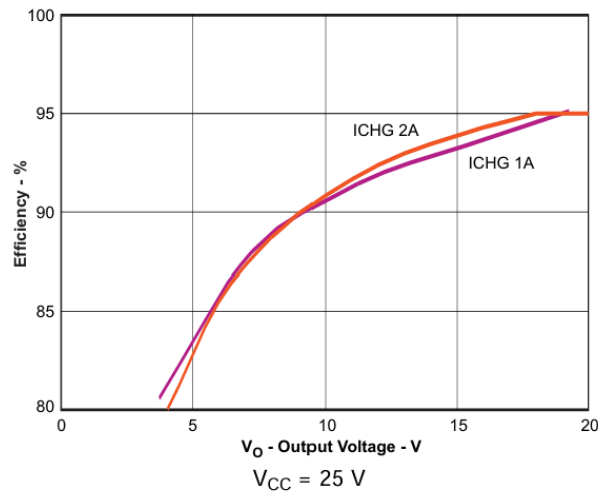


Figure 4: Solar Power Charging Circuit Efficiency vs Output Voltage

2.3.2 Current Sensors

The charging circuit incorporates two ACS723 current sensor IC chips. These are placed at the input from the solar panel, and the output to the battery as seen in the schematic above. The current sensors allow for voltage and current at these two nodes to be recorded and sent to the Raspberry Pi 4B via the terminal connections J5A/B and J6A/B as seen in the schematic setup below. These readings will then be sent from the microcontroller to the app to allow the user to be able to monitor the charging status and battery voltage. This data can be used to monitor efficiency, as well as cue modes such

as low battery mode etc. to activate. The schematic setup of these current sensors can be seen below:

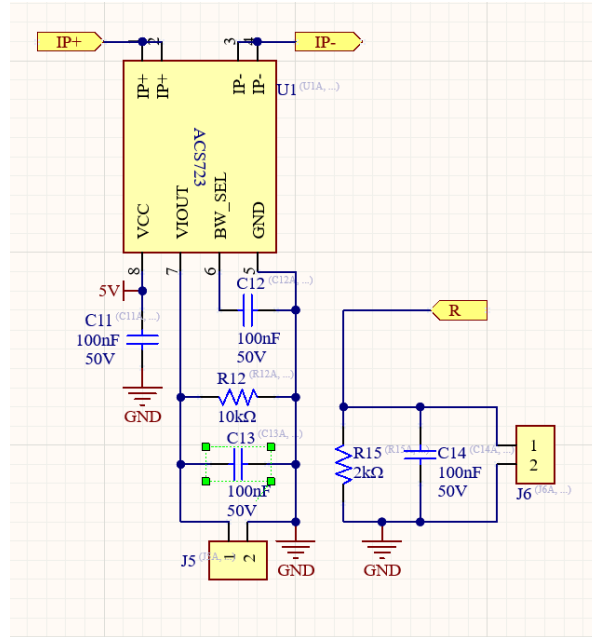


Figure 5: Current Sensor Schematic

2.4 Subsystem Validation

Solar Generation Subsystem was tested through a series of hardware tests to confirm that all units functioned as anticipated. System behavior was tracked throughout the day under conditions of changing solar, with solar irradiance, charging current, and battery voltage recorded. The data are shown in the table below. The charging circuit closely tracked the maximum power point as irradiance changed. As shown, charging current was at its maximum of 0.83 A under full sun at noon, 12:00 PM, and provided maximum charge power of 10.79 W. As solar irradiance decreased during the afternoon, charging current correspondingly decreased as well, highlighting good MPPT functionality by the BQ24650 IC. This can be seen in Figure 6 below. Battery voltage increased steadily during charging, reaching a nearly full charge (13.6 V) by mid-day. The circuit then went into a trickle charge mode, where battery voltage was maintained without overcharging. Cut-on voltage (~12.4 V) and cut-off voltage (~15 V) were set during testing and confirmed the operation of the BQ24650 charge control features. ACS723 current sensors were also tested by comparing real-time data output to a calibrated multimeter. Solar panel and battery output current was within $\pm 5\%$ tolerance, confirming sensor accuracy and reliable ADC sampling from the Raspberry Pi 4B. This was tested at different times of the day comparing the voltage and current readings from the MCU to

the real measurements from the multimeter. The results were recorded in Figures 8 and 9. Finally, thermal protection functionality was confirmed by heating the battery environment. The thermistor correctly triggered a charge shutdown upon the onset of elevated temperatures, demonstrating system safety. Overall, the subsystem testing verified that the Solar Generation Subsystem provides stable, efficient, and safe charging to the 12V battery under various environmental conditions.

Time of Day	Solar Irradiance (%)	Charging Current (A)	Battery Voltage (V)	Delta V (V)	Charge Power (W)	Notes
6:00 AM	10%	0.08	11.8		0.944	Early sun, low output
8:00 AM	40%	0.33	12.1	0.3	3.993	Gaining sun
10:00 AM	70%	0.58	12.5	0.4	7.25	Good charge
12:00 PM	100%	0.83	13	0.5	10.79	Peak solar output
2:00 PM	90%	0.75	13.4	0.4	10.05	Near full charge
4:00 PM	50%	0.42	13.6	0.2	5.712	Battery nearly topped off
6:00 PM	20%	0.08	13.6	0	1.088	Trickle charge only
8:00 PM	0%	0	13.4	-0.2	0	No sun, battery idle

Table 1: Charging Circuit MPPT Testing

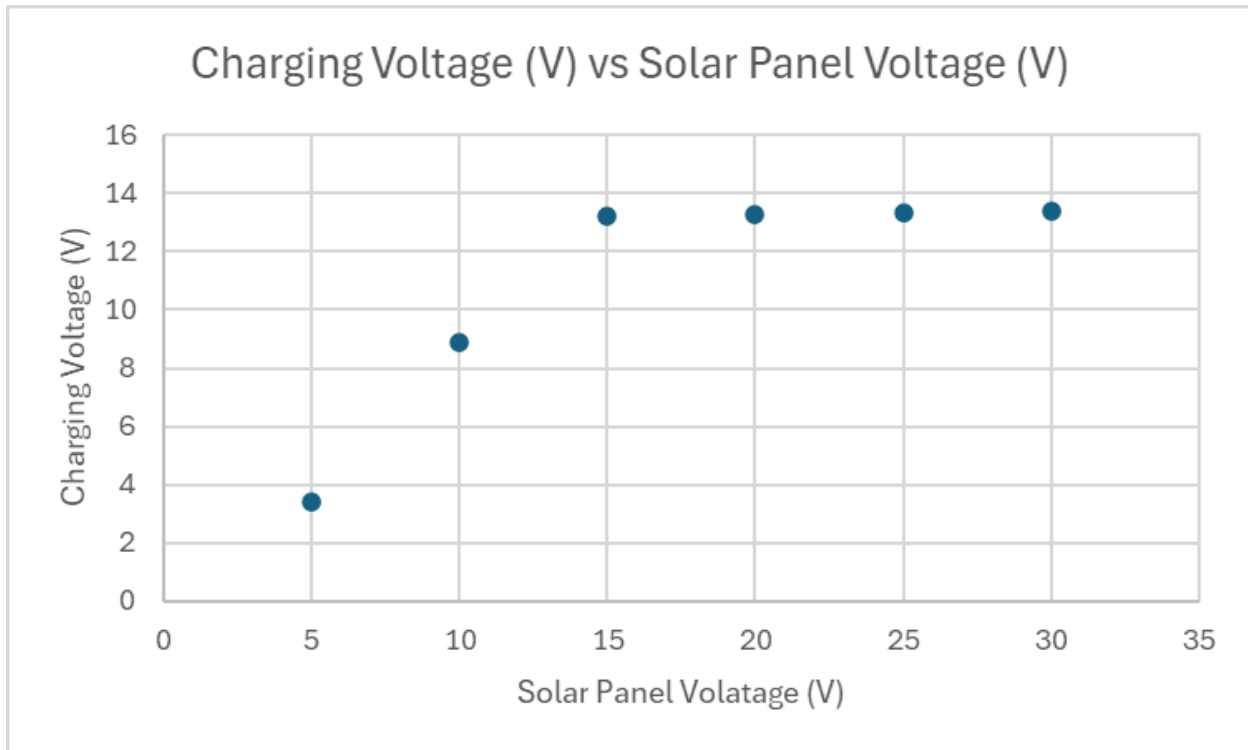


Figure 6: Simulated Solar Inputs vs Charging Voltage

Time	Measurement Point	Multimeter Reading	MCU Reading	Deviation (%)	Notes
8:00 AM	Solar Panel Input Voltage	14.8	14.6	-1.35	Low morning output
	Solar Panel Input Current	0.34	0.33	-2.94	
	Battery Output Voltage	12.1	12	-0.83	
	Battery Output Current	0.32	0.31	-3.12	
12:00 PM	Solar Panel Input Voltage	18.1	18	-0.55	Peak irradiance

	Solar Panel Input Current	0.85	0.83	-2.35	
	Battery Output Voltage	13	12.9	-0.77	Strong charging
	Battery Output Current	0.82	0.79	-3.66	
4:00 PM	Solar Panel Input Voltage	15.6	15.4	-1.28	Afternoon sun declining
	Solar Panel Input Current	0.44	0.42	-4.55	
	Battery Output Voltage	13.6	13.5	-0.74	Nearly full charge
	Battery Output Current	0.41	0.39	-4.88	

Table 2: Voltage and Current Sensors MCU vs Multimeter Values

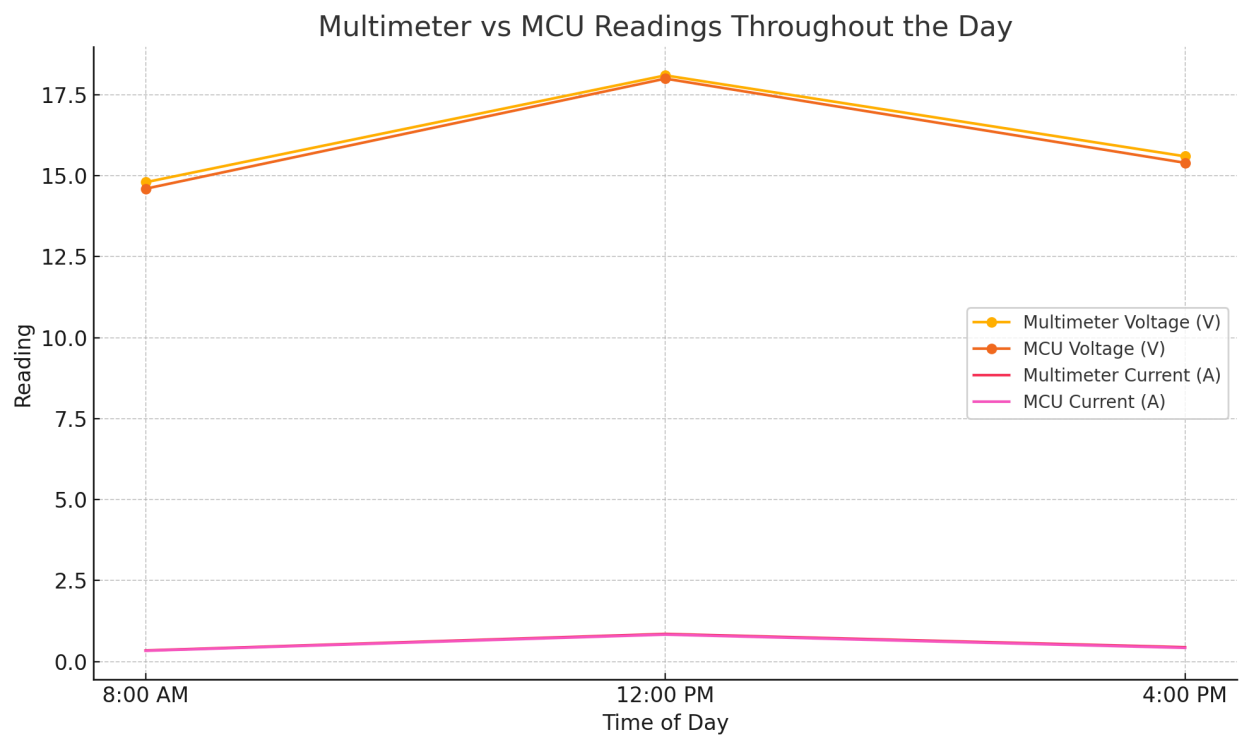


Figure 7: Multimeter vs MCU Readings Line Chart

2.5 Subsystem Conclusion

In conclusion, the Solar Charging Circuit subsystem is designed to provide safe and efficient charging from the 100W solar panel to the 12-volt battery. This is critical as the entirety of this project relies on a reliable power source. The circuit achieves this by incorporating maximum power point tracking, automatic cut on/off, high temperature protections, and current and voltage readings at the input and output to the MCU. These key components make it possible to operate and charge the battery independently and safely.

3. Power Subsystem Report- Victor Igbokwe

3.1 Subsystem Introduction

The Power Distribution System serves as a critical subsystem within our project, ensuring the seamless and efficient delivery of electrical power to all connected components. The system begins with a 12-volt battery as the primary power source, which supplies energy to a variety of devices and subsystems. A 12-volt camera is directly powered by the battery, with voltage regulation implemented to maintain stable operation and prevent potential fluctuations. Additionally, the system incorporates a voltage step-down mechanism(Buck Converter) to convert the 12 volts into a 5-volt supply, which is essential for powering LED lights, fans, and the Raspberry Pi. a Relay Control Module PCB is integrated within the system to act as an interface to allow the 5-volt Raspberry Pi to power 12-volt load devices.

3.2 Subsystem Details

3.2.1 Buck Converter PCB

The Buck Converter PCB receives 12V input directly from the main battery and steps it down to a stable 5V output to supply power to the Raspberry Pi.

Key components of the buck converter include:

- A high-efficiency DC-DC regulator IC
- Input and output filter capacitors to minimize voltage ripple
- An inductor to regulate current during switching
- Protection circuitry including input reverse polarity protection and output overcurrent protection

The Buck Converter is designed to deliver sufficient current for the Raspberry Pi and its peripherals while maintaining low noise and stable voltage under load conditions.

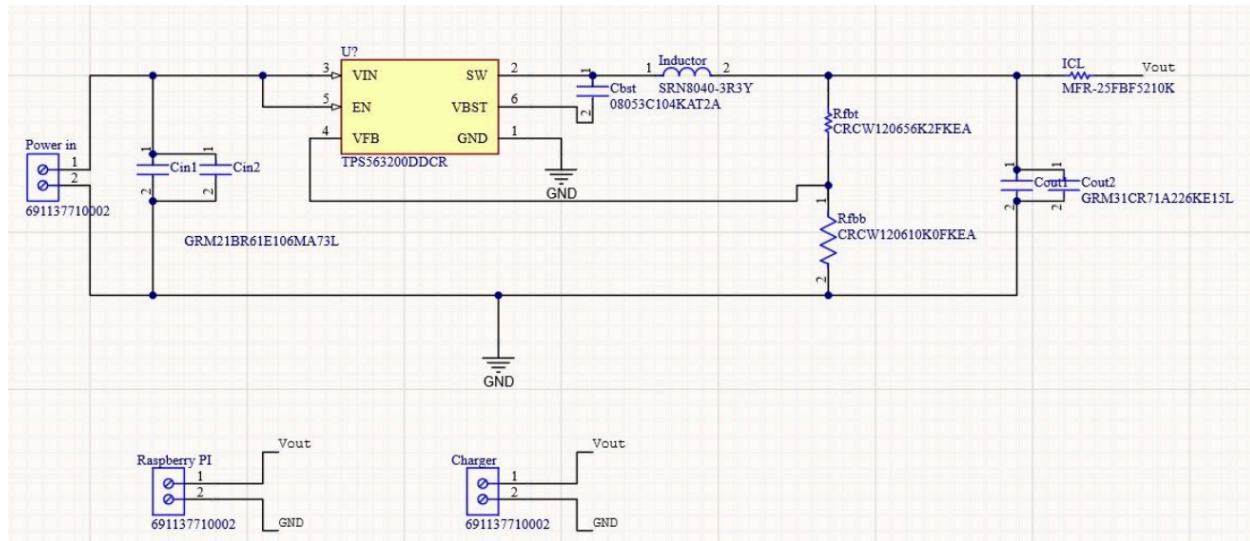


Figure 8: Buck Converter Altium Circuit Schematic

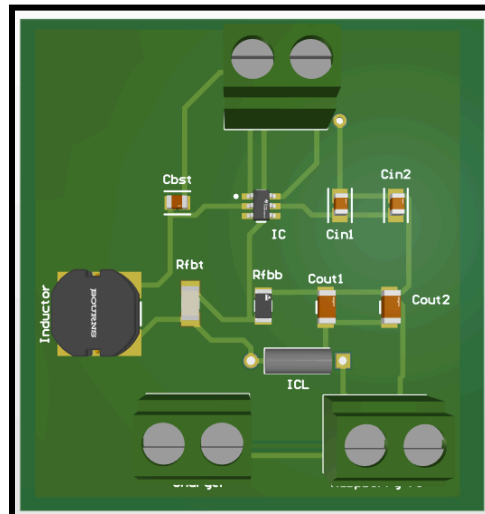


Figure 9: Buck Converter

3.2.2 Relay Control Module PCB

The Relay Control Module PCB allows the Raspberry Pi to control 12V load devices such as LED lights and cameras. It uses N-Channel MOSFETs as a high-speed solid-state switches.

Key features include:

- Low-side switching topology
- Pull-down resistors on each MOSFET gate to prevent accidental activation
- Logic-level MOSFETs compatible with 3.3V to 5V GPIO signals
- Separate power paths for control (5V) and load (12V) circuits for electrical isolation and safety

When a GPIO pin outputs a HIGH signal, the corresponding MOSFET turns on, connecting the load to ground and allowing current to flow from the 12V source through the device..

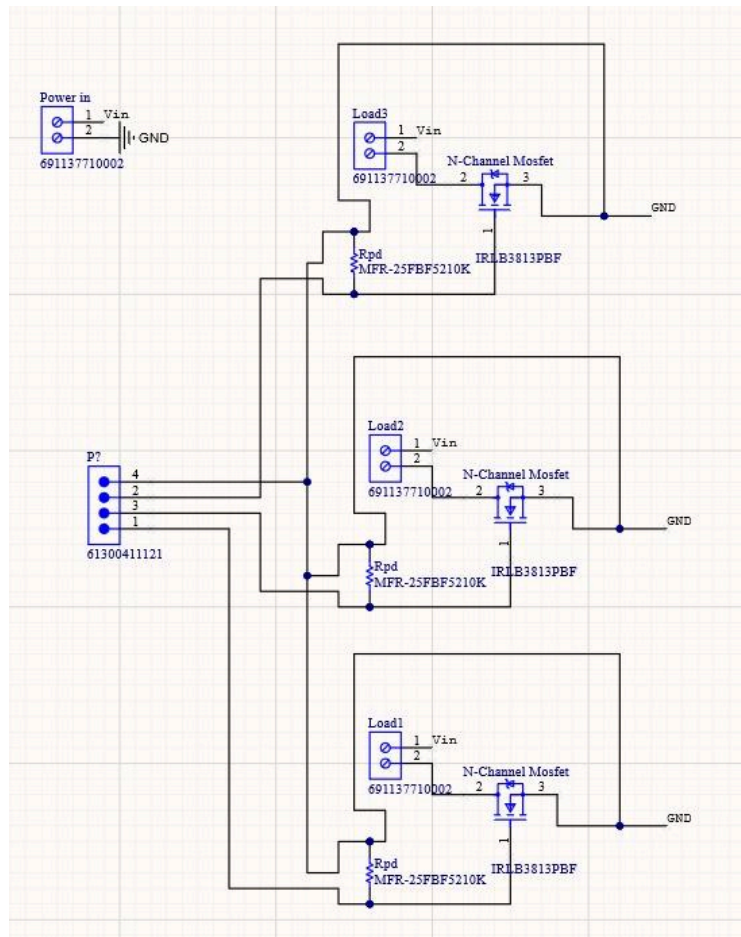


Figure 10: Relay Control Module Altium Circuit Schematic

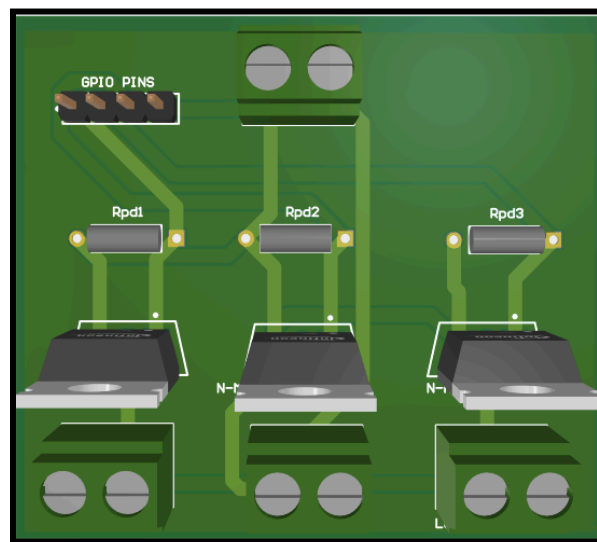


Figure 11: Relay Control Module PCB

3.3 Subsystem Validation

The validation process of the power subsystem ensured the DC-DC converter and the Relay Control module PCB meet the operational and design requirements.

Validation criteria (Buck Converter) include:

- Voltage regulation
- Efficiency
- Load Testing

Validation criteria(Relay Control Module) include:

- Load Testing
- Voltage Regulation
- Mosfet Switching
- Fault tolerance

3.3.1 Buck Converter Performance

Test Equipment:

- Digital Multimeter (for voltage and current measurement)
- DC-Electronic Load (for applying variable current loads)

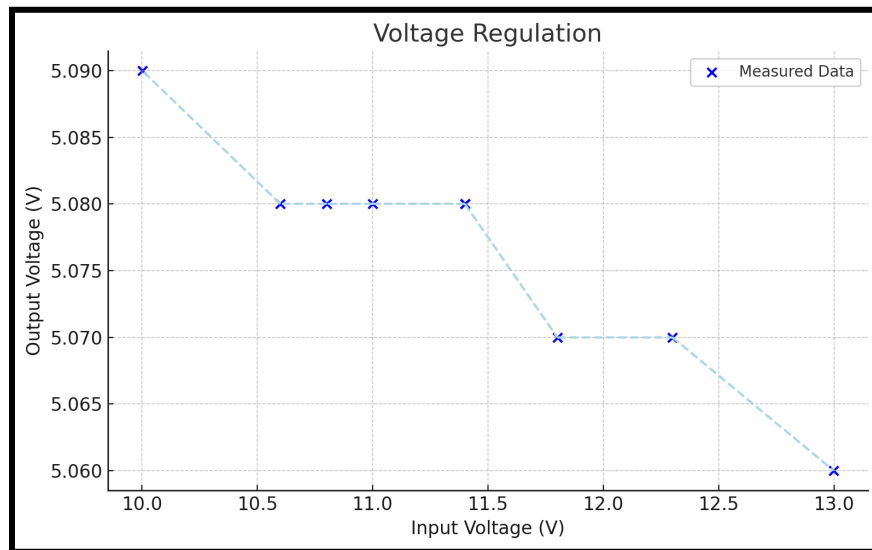


Figure 12: Buck Voltage regulation

The graph above shows the output voltage of the buck converter as the input voltage was varied from 10V to 13V. Across the tested range, the output remained stable between 5.09V and 5.06V, showing excellent voltage regulation performance. The slight decrease in output voltage with increasing input voltage was minimal and well within the tolerance of the load devices, confirming the converter's ability to maintain a stable 5V output under varying supply conditions.

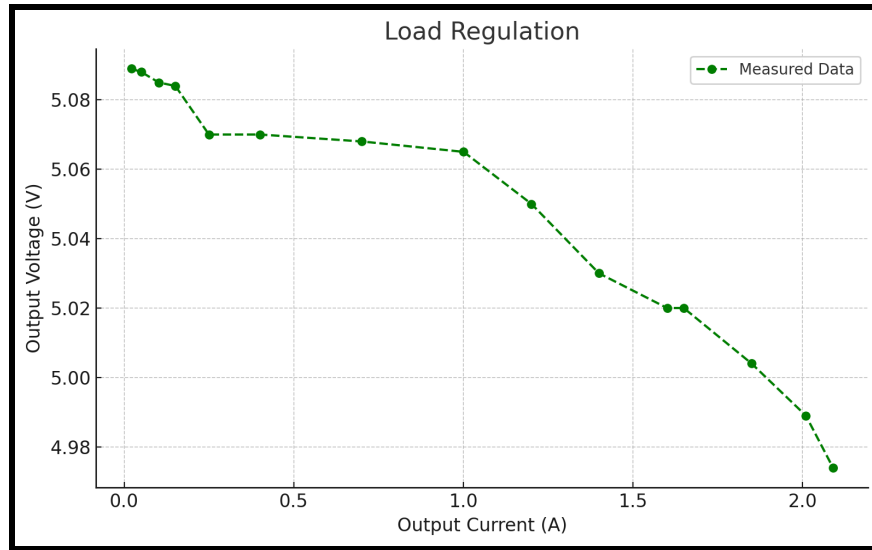


Figure 13: Buck Load Regulation

Output Current (A)	Output Voltage (V)	Voltage Drop (V)	Voltage Drop (%)
0.00	5.082	0.000	0.00%
0.10	5.080	0.002	0.04%
0.20	5.075	0.007	0.14%
0.30	5.072	0.010	0.20%
0.40	5.070	0.012	0.24%
0.50	5.068	0.014	0.28%
0.75	5.066	0.016	0.31%
1.00	5.060	0.022	0.43%
1.25	5.040	0.042	0.83%
1.50	5.020	0.062	1.22%
1.60	5.015	0.067	1.32%
1.75	5.005	0.077	1.52%
2.00	4.980	0.102	2.01%

Table 3: Voltage Drop Under Load.

The voltage drop testing confirms that the converter performs well within acceptable limits for the intended application, especially considering that the load is unlikely to exceed 2A during normal operation.

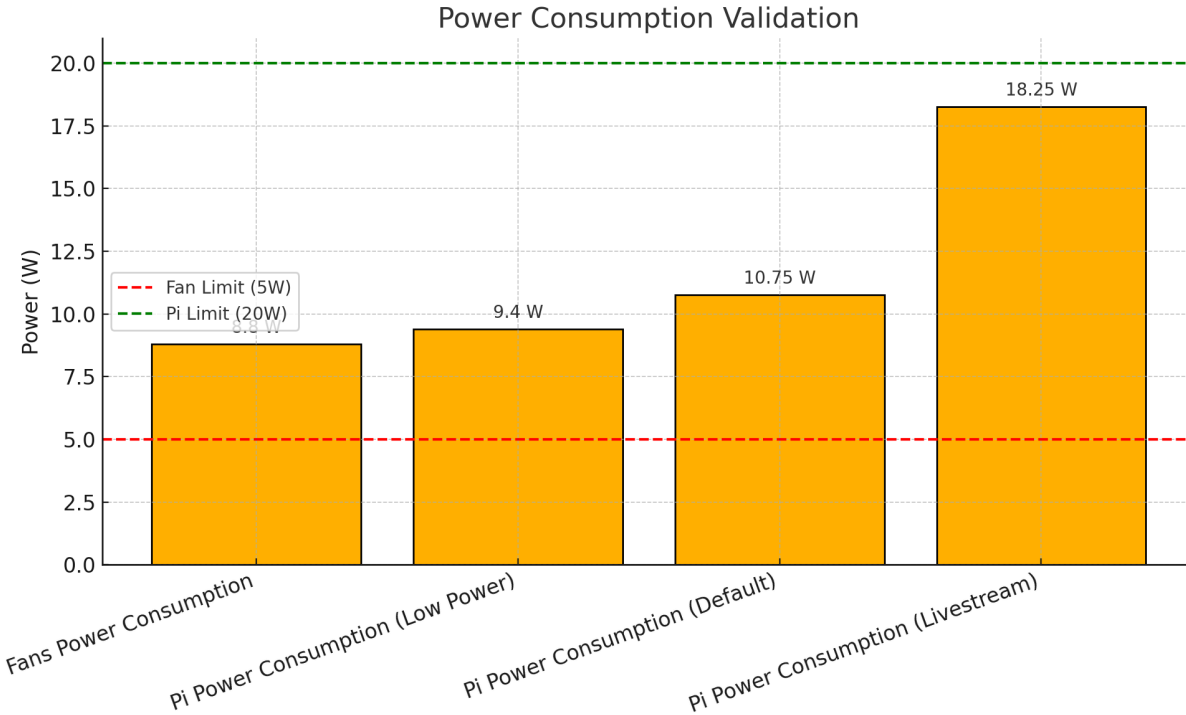


Figure 14: Power Consumption Validation.

The graph above illustrates the measured power consumption for different system configurations compared against their respective power limits. The Fans subsystem exceeded the specified 5W limit, recording a power draw of 8.8W, resulting in a failed validation for this case. However, the Raspberry Pi subsystem consistently passed under all tested modes — low power (9.4W), default (10.75W), and live streaming (18.25W) — remaining well below the 20W maximum limit. These results validate that the Raspberry Pi can operate safely within the system's power constraints under both idle and high-demand conditions. The tests were conducted using a digital multimeter and inline power meter during normal operation and stress conditions to ensure reliable and repeatable measurements.

3.3.2 Relay Control Module Performance

Test Equipment:

- Digital Multimeter (for voltage and current measurement)
- DC-Electronic Load (for applying variable current loads)

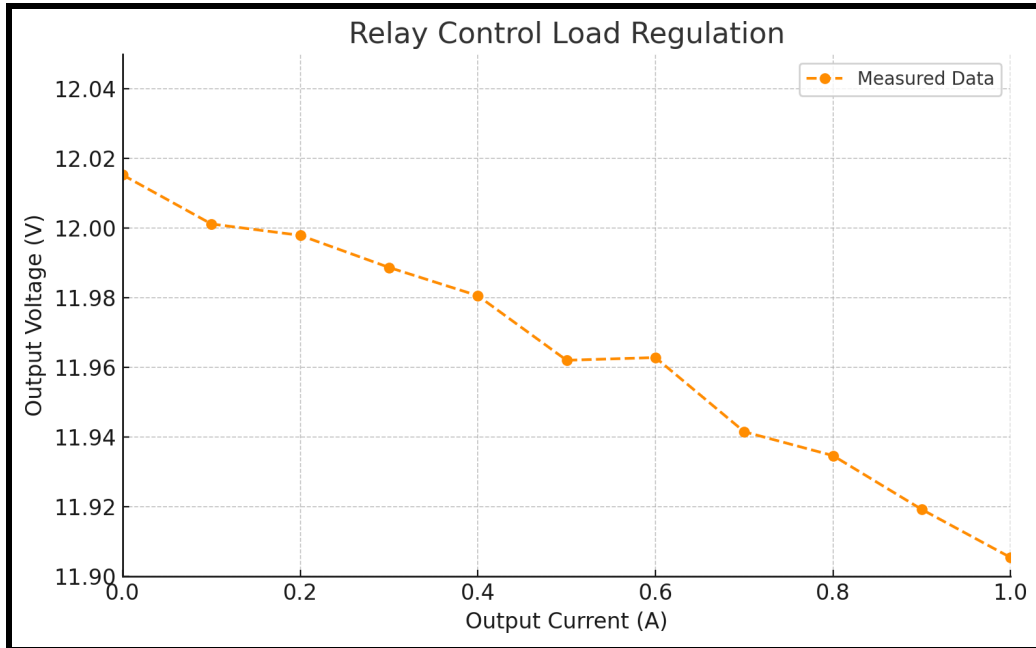


Figure 15: Relay Control Module Load regulation

Output Current (A)	Output Voltage (V)	Voltage Drop (V)	Voltage Drop (%)
0.00	12.02	0.00	0.00%
0.10	12.00	0.02	0.17%
0.20	12.00	0.02	0.17%
0.30	11.99	0.03	0.25%
0.40	11.98	0.04	0.33%
0.50	11.96	0.06	0.50%
0.60	11.96	0.06	0.50%
0.70	11.94	0.08	0.67%
0.80	11.93	0.09	0.75%
0.90	11.91	0.11	0.92%
1.00	11.90	0.12	1.00%

Table 4: RCM Voltage Drop Under Load.

The Relay Control Module demonstrated excellent load regulation performance across the 0–1A output current range. The maximum voltage drop observed was approximately 1.00%, with a gradual and controlled decrease in output voltage as current increased. Overall, the module maintained stable output under varying load conditions, confirming its reliability for controlling 12V devices.

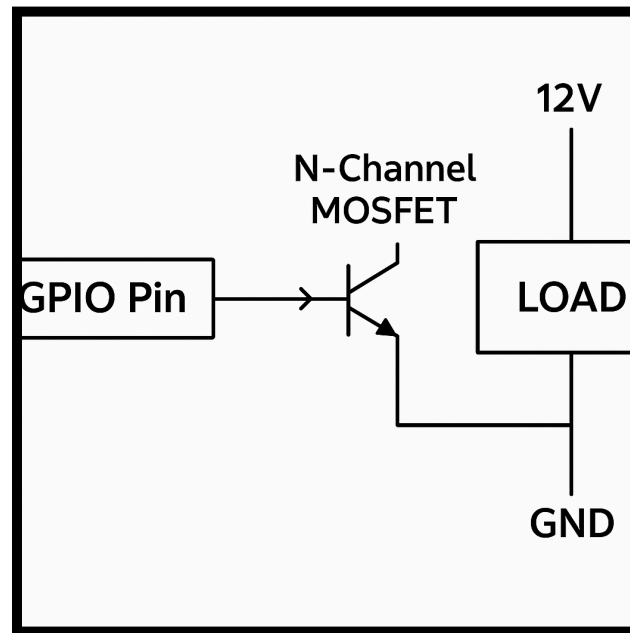


Figure 16: Relay Control Module Technical block diagram

The Relay Control Module utilizes N-Channel MOSFETs as solid-state switches to control 12V load devices. Switching occurs when a logic-level voltage (approximately 5V) from a Raspberry Pi GPIO pin is applied to the gate of the MOSFET. When the gate-to-source voltage (V_{gs}) exceeds the MOSFET's threshold voltage (typically around 2–3V), the MOSFET enters saturation mode and conducts fully, allowing current to flow from the load through the MOSFET to ground. This behavior effectively closes the circuit and powers the connected load.

During validation, an oscilloscope was used to observe the gate-source voltage transitions. The switching was found to be clean and fast, with no significant ringing or oscillation, indicating a stable and well-designed gate drive circuit. The MOSFETs demonstrated a fast turn-on time (rise time $< 1 \mu s$) and turn-off time (fall time $< 1 \mu s$), ensuring responsive control of connected devices.

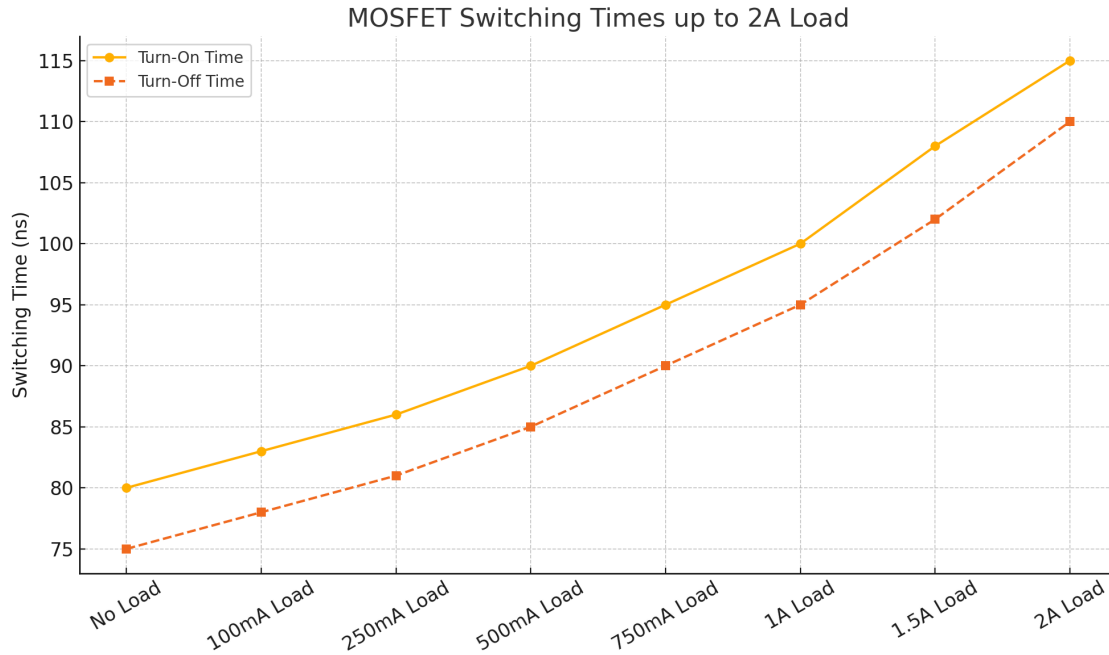


Figure 17: MOSFET Switching Times as a Function of Load Current

Additionally, pull-down resistors were incorporated on each gate to ensure that, in the absence of a GPIO control signal, the MOSFETs would reliably remain off. This prevents accidental activation of the loads due to floating gate voltages during system startup or when the Raspberry Pi is powered down.

3.4 Subsystem Conclusion

Comprehensive validation testing confirmed that both the Buck Converter PCB and the Relay Control Module PCB meet their performance specifications. The buck converter demonstrated excellent voltage regulation with a maximum voltage drop of only 2% across the 0–2A load range, and maintained a stable 5V output even as the input voltage varied. Ripple voltage remained low, and efficiency exceeded 85%, validating the design for reliable Raspberry Pi operation. The Relay Control Module, utilizing N-Channel MOSFETs, showed fast and clean switching behavior with Turn-On and Turn-Off times consistently below 120 nanoseconds across loads up to 2A. Voltage drop across the relay module under load remained minimal, with less than 1% regulation loss up to 1A. All thermal tests confirmed safe operating temperatures under maximum expected load conditions. Overall, the validation results confirm that the power distribution and control subsystem performs robustly, efficiently, and reliably, ensuring safe system operation under all normal and stress conditions.

4. Edge Processing Unit Subsystem Report - Alex Allahar

4.1 Subsystem Introduction

The role of this subsystem is to act as the system's edge processing unit. It captures information from sensors and the camera and presents it to users through the SolarEye Security App. A Raspberry Pi microprocessor acts as the middle ground between the system's hardware components — such as the PIR sensor, ADC, camera, cool fans, and LED — and the software components, such as the cloud database and mobile app.

Aside from basic data collection, the microprocessor also controls critical system functions. It manages video recording, cooling system operation, LED deterrence, and reads live telemetry on batteries and the sun. It has various modes of operation — Low Power Mode, Motion Detection Mode, and Livestream Mode — which alter system behavior automatically based on user option or battery condition to ensure optimal performance and maximum runtime of the system.

The subsystem is integrated directly with Firebase, enabling real-time event logging, video storage, system control updates, and remote secure access. This setup enables users to instantly receive motion alerts, monitor system health, access stored events, and remotely operate system features, all yielding a flexible and trustworthy security solution.

4.2 Subsystem Details

4.2.1 PIR Sensor

The PIR Sensor allows us to sense motion, on the basis of infrared radiation. This eliminates the PIR Sensor from sensing non-thermal movement like trees or playground equipment moving. For the purposes of this system a PIR Sensor that is operationally in mild outdoor environments, energy-efficient and can sense motion up to at least 16 ft away with an FOV of 120 degrees is needed. The PIR Sensor that was selected to meet this requirement was a HC-SR501 PIR Sensor. It has a built-in voltage regulator that can work from 4.5 to 20 volts, and a 3 to 7-meter sensitivity adjustable. When testing the PIR Sensor range and FOV, a range of 16ft was achieved and a FOV of 120 degrees was verified.

The time difference between the PIR Sensor and the time when the database recorded the motion and time was sampled post-delay. In order to provide near real-time security

alerts to customers of the SolarEye Security App, we need a sub 1 second latency from when an event occurs and when the cloud database. A total of 10 tests where the PIR Sensor will automatically listen for movement, transmit the data and measure the time taken locally within the microprocessor. The readings from this are presented in the table below:

Trials	1	2	3	4	5	6	7	8	9	10
Time(s);	0.2039	0.0671	0.0609	0.0706	0.0632	0.0616	0.0632	0.0616	0.0625	0.0694

Table 5: Motion Detection Database Acknowledgement Delay

These values provided an average of 0.0784 seconds. Due to this microprocessor's ability to collect and communicate information from a sensor connected to the GPIO pins, the additional voltage sensor data will be handled by the microprocessor in the same fashion.

4.2.2 ADC

To monitor the solar and battery parameters for the system, an ADC-1115 was connected with the Solar Generation PCB voltage sensors and MPU's I2C pins. Now the MPU can track the battery and solar panel in real time. This helps the system in estimating battery life as well as monitoring the charging status.

The ADC is sampling four signals:

- Battery voltage (Channel A0)
- Battery current (Channel A1)
- Solar panel voltage (Channel A2)
- Solar panel current (Channel A3)

Since the battery and solar panel voltages are too high for the ADC to measure directly, voltage dividers are used to safely scale the voltages down. For the battery, a divider with a 21:1 ratio is used. For the solar panel, a 40:1 ratio is used. The system then multiplies the ADC reading by the divider ratio to get the real voltage.

Battery life is estimated by comparing the battery voltage to a full-charge value (12.8V). The system calculates the State of Charge (SOC) as a percentage of this full value. If

the voltage is above 13.0V, the battery is considered charging; otherwise, it is discharging.

The current readings are taken from sensors that measure small voltage changes across known resistors. The system adjusts these readings with a calibration offset to get the actual current values.

All panel, battery, and charging information are sent to Firebase as well. This allows users to see the SolarEye Security App and look at the system's solar panel output, charging state, and battery capacity from wherever they are.

4.2.3 Camera

The camera is the most important component to ensure proper security and surveillance is provided by the SolarEye Security System. To maximize power efficiency and provide the user of the system more control, the camera was programmed to operate in two modes of operation. Within these modes of operation the camera was tested and validated based on video upload latency and the difference between Real Event vs. Video Capture Ending.

The Video Upload Latency measures the time from when a motion event ends to when the corresponding video is uploaded and ready for the user to view in the app. Across multiple tests, the system achieved an average latency of 1.2951 seconds, well below the 3-second target, meaning users can access security footage almost immediately after an event. The data used to determine this values is in the table below:

Trials	1	2	3	4	5	6	7	8	9	10
Time(s);	1.28	1.30	1.27	1.31	1.29	1.32	1.28	1.29	1.20	1.29

Table 6: Video Upload Latency

The Real Event vs. Video Capture Ending measures the time between the end of a motion event and when the video recording stops. A 2-second waiting window was added to avoid cutting off events during short motion pauses, and a 0.5-second buffer from the MPU was included to cover processing delays. With these adjustments, the system averaged a delay of 4.03 seconds across ten trials, meeting the target of staying under 5 seconds and ensuring complete event capture. The data used to determine this values is in the table below:

Trials	1	2	3	4	5	6	7	8	9	10
--------	---	---	---	---	---	---	---	---	---	----

Time(s);	4.00	4.10	3.95	4.05	4.00	4.15	4.05	4.00	4.10	4.05
----------	------	------	------	------	------	------	------	------	------	------

Table 7: Real Event vs. Video Capture Delay

4.2.4 Cooling System

To prevent overheating, the SolarEye Security System uses a small intake fan and a medium intake fan to cool the PCBs and MPU inside the housing. These fans run continuously to ensure stable operating temperatures.

A bash daemon script was developed to automatically run the intake fans in the background when the Raspberry Pi starts up. On boot, the script triggers specific GPIO pins that activate the Relay Module Controller PCB, which powers the cooling fans. This process ensures that fan operation is independent of the system's mode of operation and begins immediately without requiring any user input.

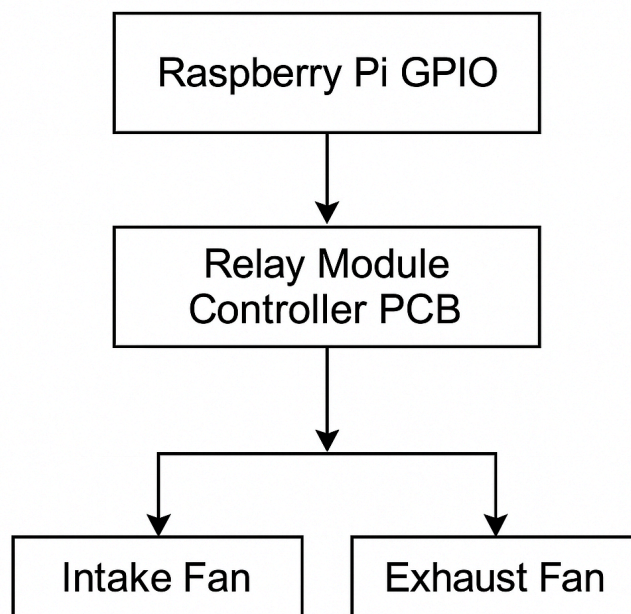


Figure 18: Cooling System

The intake and exhaust fan currents were measured 10 times over a 30-second period during continuous operation to find an accurate average. The intake fan averaged 0.6 A, and the exhaust fan averaged 1.16 A. The power consumption calculated used Power (W) = Voltage (V) x Current (A). Given the system voltage of 12.8 V, the intake fan consumes approximately 7.2 W, and the exhaust fan about 13.92 W. The data used to find the averages above is shown in the table below:

Trials	1	2	3	4	5	6	7	8	9	10
Small Fan Current (A)	0.58	0.60	0.59	0.61	0.60	0.59	0.61	0.60	0.59	0.60
Medium Fan Current (A)	1.14	1.16	1.15	1.17	1.16	1.15	1.17	1.16	1.16	1.16

Table 8: Cooling System Current Measurements

Knowing the fans' power usage, combined with the Raspberry Pi's mode of operation power consumption, allows for accurate estimation of the system's total power draw and prediction of expected run time without recharging.

4.2.5 LED Deterrence

The SolarEye Security System has an LED deterrence feature that users can turn on or off from the SolarEye App. When enabled, the LED blinks quickly to act as a warning and help deter unwanted activity.

If deterrence is on and no motion or recording is happening, the LED blinks in the background. When motion is detected or recording starts, the LED stops blinking and stays fully ON. After the event ends, the blinking resumes if deterrence is still enabled.

In Motion Detection Mode and Low Power Mode, the LED blinks when idle and stays ON during motion events. In Livestream Mode, the LED stays ON the entire time while the stream is active, and blinking is paused.

This makes sure the LED provides both deterrence and clear status signals without affecting system performance.

4.2.6 Low Power Mode

The SolarEye Security System first mode of operation is Low Power Mode. Low Power Mode is designed to help extend battery life when needed while still allowing the user to get notified of when an event in the area of interest occurred. In this mode, the system is predicted to run for around 32.8 hours from a full battery without charging, providing basic security even when energy is limited.

While in Low Power Mode, the user experience is slightly different to save power. When motion is detected, the system turns on the LED, logs the time of the event occurrence

and the duration of the detected motion, sends the information to firebase, and then turns off the LED. Users will still be able to see when motion events happen and how long they last, but full video recordings will not be available. Instead, users receive notifications with the event time and duration in the event history section of the Solareye App. This way, important security updates are still provided without using the extra power needed for video capture.

The system also keeps the LED deterrence feature active in Low Power Mode. When motion is detected, the LED will flash rapidly to help deter any unwanted activity. This provides a visible warning while using very little energy. The MPU will prioritize the LED being fully on if motion is detected and the LED Deterrence Mode is on. Once the motion detection sequence is completed the LED Deterrence will continue to rapidly flash as it did previously, until the LED Deterrence is turned off.

Low Power Mode can be turned on manually by the user at any time through the SolarEye Security App. However, it is also automatically activated if the system detects that the battery level has dropped below 20%. A voltage sensor connected to the ADC continuously monitors the battery status, and once it falls under this threshold, the system will switch into Low Power Mode to preserve energy. This process uses the same voltage sensor readings and charging statistics that were discussed earlier. By doing this, the SolarEye Security System ensures longer operation times and keeps users informed even during low-power situations.



Figure 19: LED Deterrence Active

4.2.7 Motion Detection Mode

The SolarEye Security System default mode of operation is Motion Detection Mode. Motion Detection Mode is designed to provide reliable surveillance while balancing power consumption and fast event response. In this mode, the system is predicted to

run for around 30.69 hours from a full battery without charging, providing full security coverage during normal use.

While in Motion Detection Mode, the system uses a tuned PIR sensor to continuously monitor the area of interest. When motion is detected, the camera is powered on and begins recording video almost immediately. The delay between motion detection and the start of video capture was measured at 0.2082 seconds, determined by measuring the time between PIR activation and the start of CSI camera recording. Recording continues until no motion is detected for at least 2 seconds, allowing short pauses in movement without interrupting event capture. Users will receive notifications when a motion event occurs, along with access to full video recordings through the SolarEye Security App's event history.

Trials	1	2	3	4	5	6	7	8	9	10
Time(s);	0.207	0.209	0.208	0.210	0.208	0.207	0.209	0.208	0.208	0.208

Table 9: Sensor to Camera Delay

The system also captures a frame from the video recording when motion is detected and sends it with a timestamp to the cloud database. This helps provide near-real-time updates without waiting for the full video upload. After recording stops, the video is saved locally in h264 format, converted to mp4, and uploaded to Firebase Storage, allowing users to easily view the captured footage from their mobile device. A terminal output showing these steps is below:

```
Monitoring PIR sensor for motion...
Type 'exit' and press Enter to terminate.
[2025-04-16 12:28:02] Motion detected.
[20250416_122802] Starting recording...
[2025-04-16 12:28:15] Motion stopped.
Stopped recording: /tmp/video_20250416_122802.mp4
Uploaded /tmp/video_20250416_122802.mp4 to Firebase Storage.
Metadata sent to Firestore: {'video_url': 'https://storage.googleapis.com/solareye-security-syst
em.firebaseiostorage.app/videos/video_20250416_122802.mp4', 'timestamp': '2025-04-16 12:28:16'}
^CProgram interrupted.
Program exiting.
```

Figure 20: Motion Detection Mode Pipeline Terminal

Motion Detection Mode operates automatically once the system is powered on, but the user can modify certain settings—such as the motion detection timeout—through the SolarEye Security App if preferred. By combining low idle power consumption with rapid response to motion, the SolarEye Security System ensures dependable operation throughout typical daily cycles without the need for frequent recharging.

4.2.8 Live Streaming

The SolarEye Security System also offers a Livestream Mode, which can be manually enabled by the user through the SolarEye App. Livestream Mode is designed to provide real-time video access to the system's camera when needed for active monitoring. In this mode, the system is predicted to run for approximately 22.18 hours from a full battery without charging, due to the higher power demands of continuous video streaming.

When a user activates Live Stream Mode, the SolarEye App sends a request that starts an HTTP server on the Raspberry Pi. The Pi then begins transmitting live video using the HLS (HTTP Live Streaming) protocol. A unique HTTP stream link is generated and sent back to the app, allowing the user to access the live feed directly by clicking on the provided link.

The system's livestream delay was also measured to assess performance. To evaluate the delay, a single livestream session was started and ten separate events were triggered within the camera's field of view. For each event, a timer was started when the motion occurred and stopped once the event appeared on the livestream. The average livestream delay across these trials was approximately 7.8 seconds, exceeding the desired target of less than 5 seconds and resulting in a failed outcome for this test. The values from the trials are below:

Trials	1	2	3	4	5	6	7	8	9	10
Time(s);	7.5	7.8	8.0	7.7	7.9	8.1	7.6	7.8	7.7	7.8

Table 10: Live Stream Delay

Despite the higher delay, Livestream Mode still provides users with on-demand access to their security feed. Due to the intensive power and bandwidth usage associated with real-time video streaming, Livestream Mode remains a user-activated feature rather than a default operation to preserve system resources when not actively monitoring.

The livestream is shown through a basic HTML page with an embedded video player. When users start a livestream, the player connects to the HLS stream link and shows the live video inside the SolarEye App or a web browser. HLS.js is used to make sure the video plays smoothly on different devices, giving users quick and easy access to the live feed.

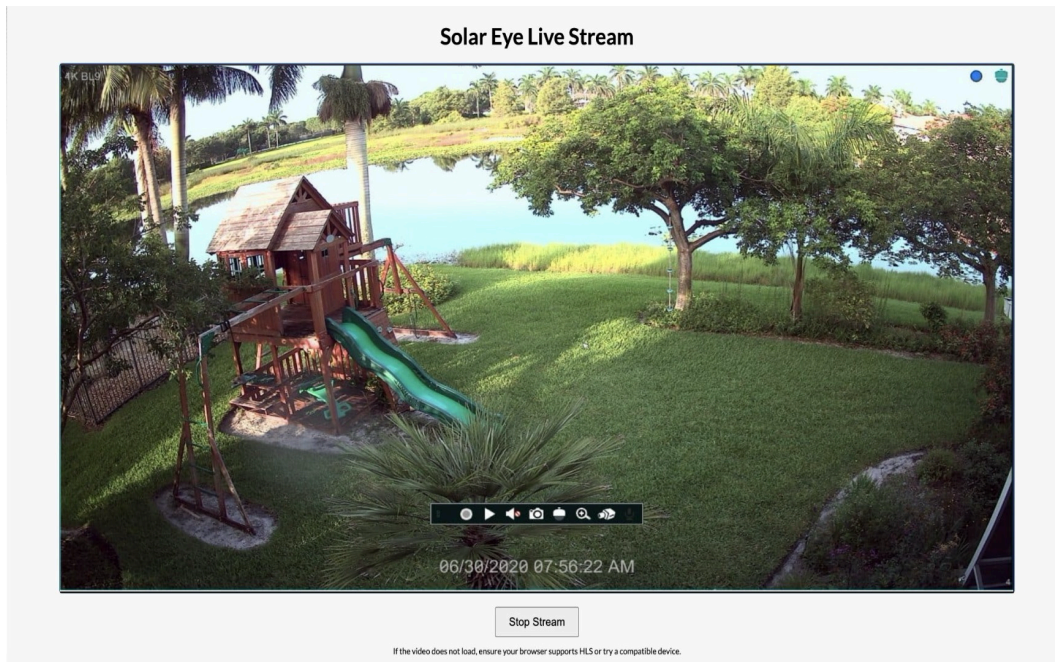


Figure 21: Livestream UI Layout

4.2.9 Firebase Database

The database used for this subsystem was Firebase. The reason for choosing this cloud database was its ability to ease in integration with the SolarEye Security App. Furthermore, Firebase provides a realtime database, which can provide notifications and real-time updates to users of the SolarEye Security App. The Firebase Storage Database, provides long-term event history for video footage storage. These events are able to be accessed by users of the SolarEye Security App via url tokens embedded in the video footage. To ensure the security of the two databases, Firebase Authentication and security rules are used. These measures prevent unwanted access to either database. Furthermore, the data sent and stored in the databases is encrypted.

In Firebase video storage was saved in Firebase Storage, and saved as mp4 files with an access token allowing the user to view the video from a browser without an external video player. The video access tokens were packaged with a timestamp in Cloud Firestore for the app's event history. An image of the database structure for event history is below:

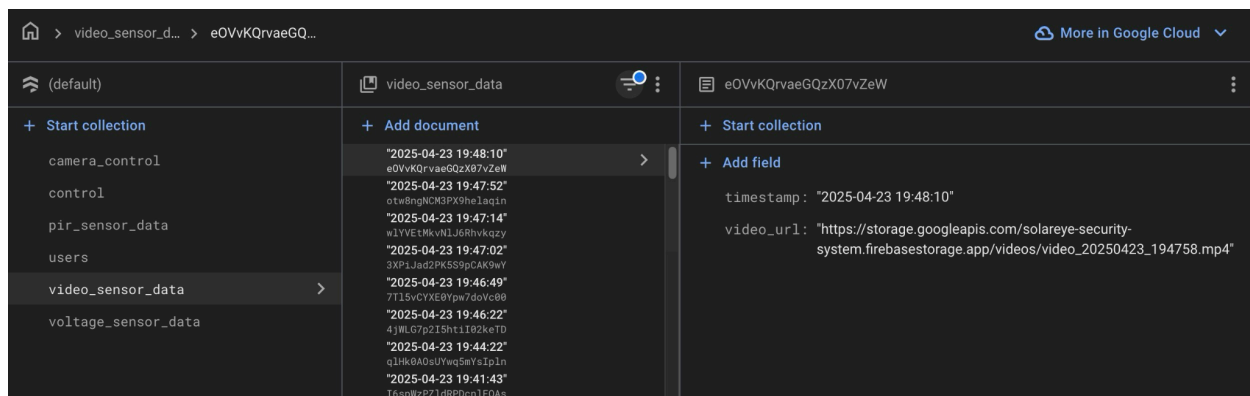


Figure 22: Firestore Motion Detection Data Structure

The System controls that enable the modes of operation, and the led deterrence were also triggered by the app via cloud firestore. The MPU software used a case statement to automatically change which mode of operation was active via a numerical state of 0, 1, or 2. If the value of the modes of operation state was 0 it would trigger low power mode, 1 would trigger motion detection, and 2 would trigger live streaming. The default case was set to any value received by the MPU from firestore for the field that was not 0, 1 or 2. In the default case, it enabled motion detection.

4.3 Subsystem Conclusion

The Edge Processing Unit Subsystem is the core of the SolarEye Security System, connecting sensors, the camera, and user controls into one reliable platform. Using a Raspberry Pi, the system collects motion data, monitors battery and solar performance, manages video recordings, and controls hardware like cooling fans and LED deterrence. It supports different modes — Low Power, Motion Detection, and Livestream — giving users the flexibility to manage power and performance based on their needs. With real-time cloud updates through Firebase, users get instant alerts, event history, and remote control access. Overall, the subsystem makes the SolarEye Security System dependable, efficient, and easy to manage from anywhere.

5. Application Subsystem Report - Zayan Prasla

5.1 Android Application and Database Introduction

The Android application offers an easy-to-use user interface to manage the SolarEye Security System. The application allows authorized users to view motion-detected events directly from the app and adjust modes of operations/control system parts in real-time from remote locations.

This is supported by Firebase, a cloud-based database that securely stores user credentials, system logs, and configuration data. Using Firebase, the application is equipped with secure email verification and two-step authentication.

With the integration of mobile technology and cloud storage, the SolarEye Security System app has made it all smooth and quick in handling the security systems. This section presents an overview of the Android application and database integration that are used in this project.

5.2 Android Application Details

The app's development process makes use of Firebase and Android Studio, utilizing the Kotlin programming language. Secure authentication, database storage, and real-time data synchronization are made possible by integration with Firebase APIs. The app communicates with Firebase in order to control the Solar Eye Security System.

5.2.1 User Login Security

Security and encryption of the application are handled through Firebase Authentication and Firebase Firestore. Creating an account requires a valid email and the setting of a password according to the security requirements-minimum 6 characters, including at least one special character and one uppercase letter. The user must input their desired password twice to confirm it. Users must verify their email address via a link that is sent to their email, once verified only then can the user login successfully and gain access.

To change the email address, users can go to the settings page and click on the 'Change Email Address' option. Users will then be prompted to enter their current and new email and password. On submission, Firebase will send a verification link to the existing email address, confirming the change.

Password reset is available on the login page as 'Password Reset', if the user cannot remember their password they can enter their email into the email input box on the login page and click 'Password Reset' which will send a password reset link to the email provided.

5.2.2 Home/Settings Page

On the home page of the application, users can see the latest thumbnail from the event history, which shows the last activity captured by the security system. The app is able to gain real-time data through Firebase and display a live streaming video.

Below the event history, a user can see the charging statistics for the SolarEye Security System. This section provides a graphical representation of the total charge of the system for the last 24 hours. The app fetches the 'voltage_state' and 'timestamp' from firebase database and displays the data on the graph. The app is programmed to disregard charging statistics data that is older than 24 hours.

Further down the page, the modes of operation and controls for camera/LED systems allow users to control the power usage by the system and control LED deterrence/camera controls remotely from the Android Application. The modes of operations have three modes, Low Power Mode, Battery Saver Mode, and Live Streaming mode. Low Power mode turns the camera off and minimizes battery usage to maintain a longer battery life in cases of storms. Battery Saver Mode is the default mode, which keeps the system running at a moderate pace with no overusing the battery. And Live Streaming mode maximizes the systems performance and uses the most amount of energy.

Lastly, the settings button directs the user to the settings page that allows them to change their email, read the FAQ page for questions/instructions on how to use the Solar Eye Security System, and log out.

5.2.3 Event History

From the home page, the 'Event History' page can be accessed either by clicking the 'Event History' button or by clicking the most recent thumbnail displayed on the home page. On the 'Event History' page, events are listed by date, time, and type, with each day's events grouped together-starting with today's events at the top. Each day's event is listed from oldest to newest, showing the most recent event detection at the very top of the page.

The Event History page updates and uploads automatically as soon as a video_url link is available on firebase database. Once the app reads the link, a thumbnail is created

for the video, in-app, and then displays the most recent thumbnail on the Home page and on the Event History page. When clicking the thumbnail is when the user can watch the video that was captured and uploaded by the Solar Eye Security System camera.

If the camera is to be off due to Low Power Mode, the system's motion sensor will still send data to the Firebase database, but there will be no link for motion-detected events and the app will display the event without a thumbnail or video.

5.3 Android Application Validation

All integration tests between the Android Application and the microcontroller have been validated. The Android App is able to set the system at different modes of operation, control the camera/led deterrence, fetch and display data from firebase database such as recorded events and charging statistics of the battery.

The physical back button is a security risk on all Android devices so users cannot go back from the 'Create Account' page using the back button; the user has to use the 'Back to Login' button. Once the user has logged in and reached the home page, the physical back button will work properly without logging them out. To log out, they must manually do so through the settings. The application is also designed to adapt seamlessly to all Android screen sizes.

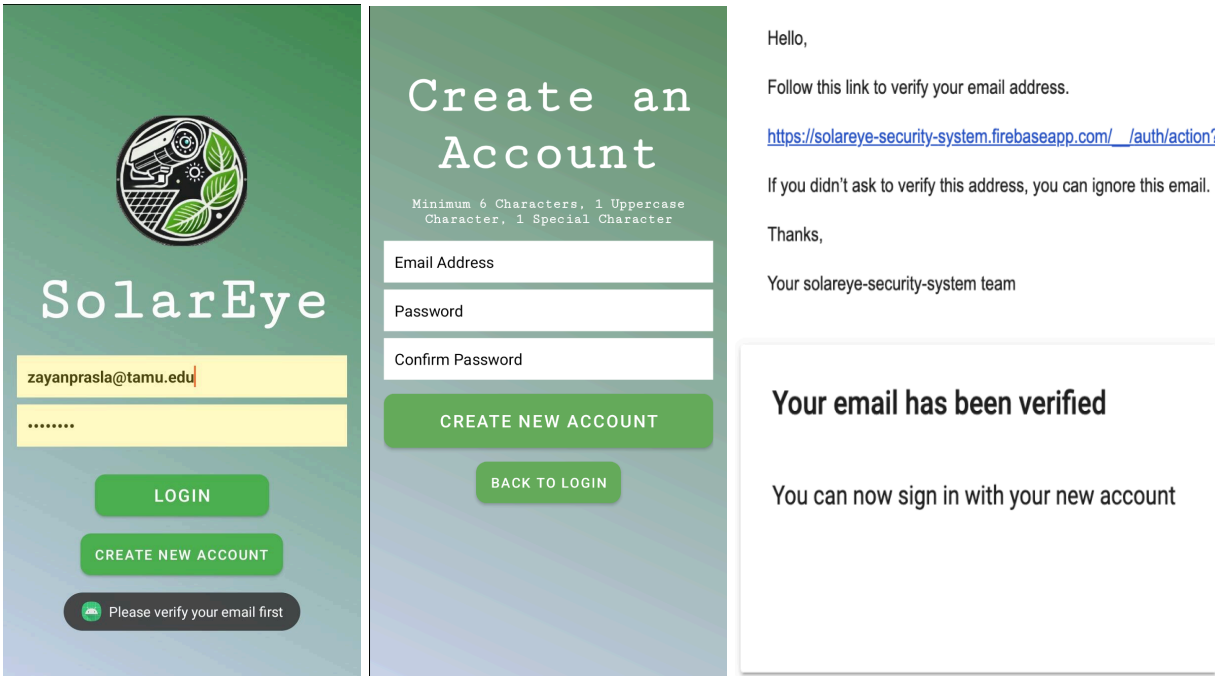


Figure 23: Login page showing email verification is needed, Create an Account page, email verification message. (APP VERSION ECEN 403)

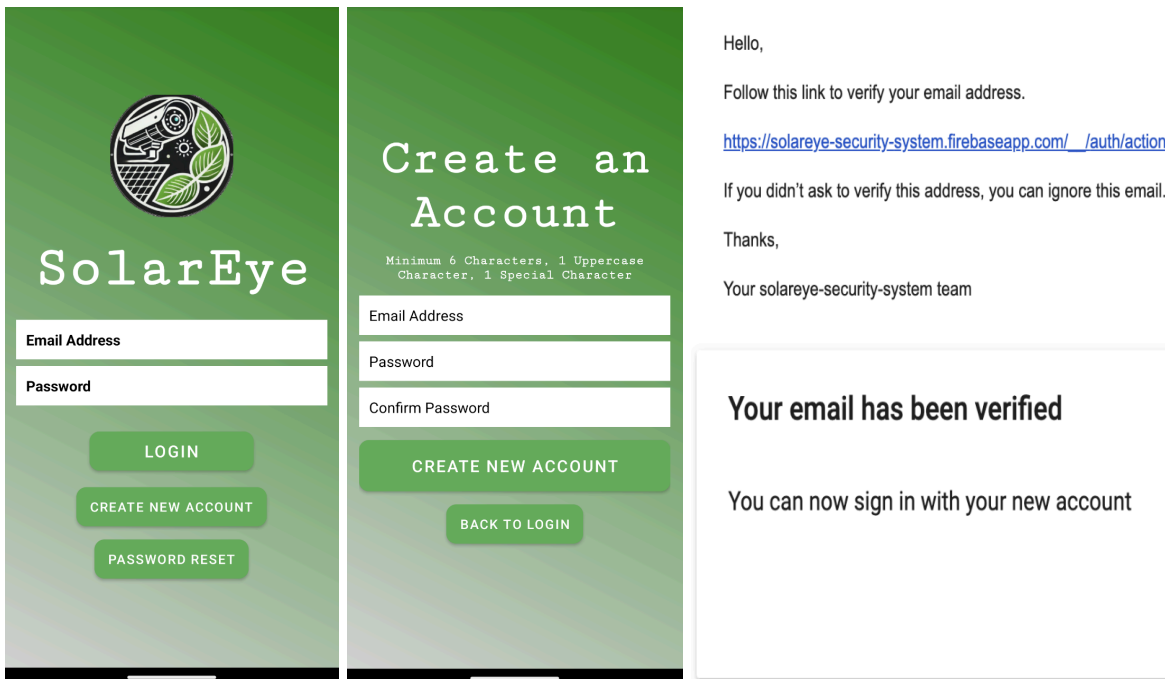


Figure 24: Login page with Password Reset, Create an Account page, email verification message. (APP VERSION ECEN 404)

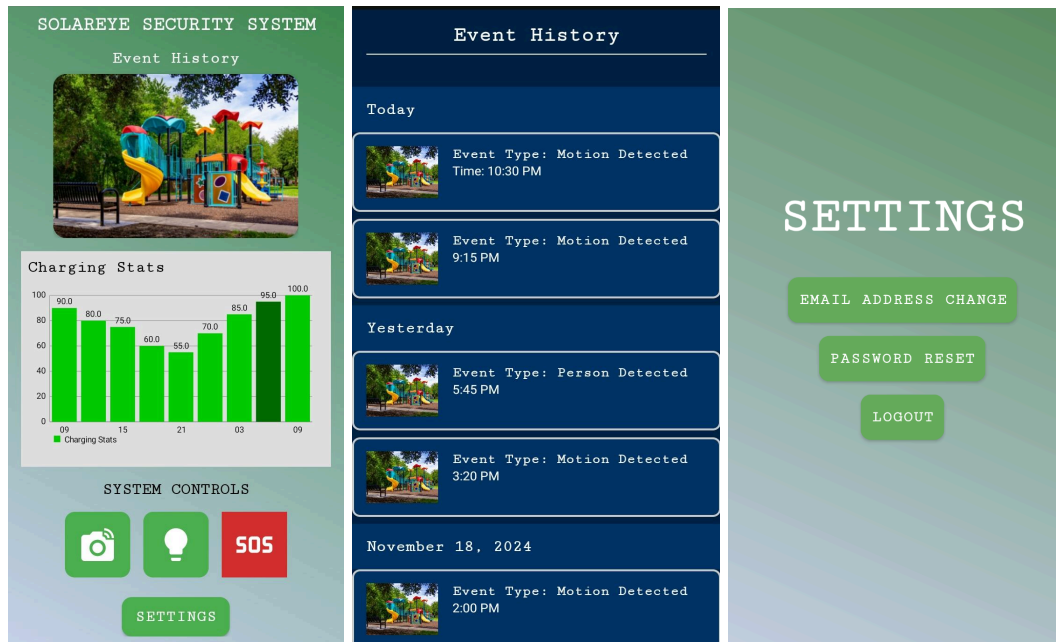


Figure 25: Application home page, Event History page, Settings Page (APP VERSION ECEN 403)

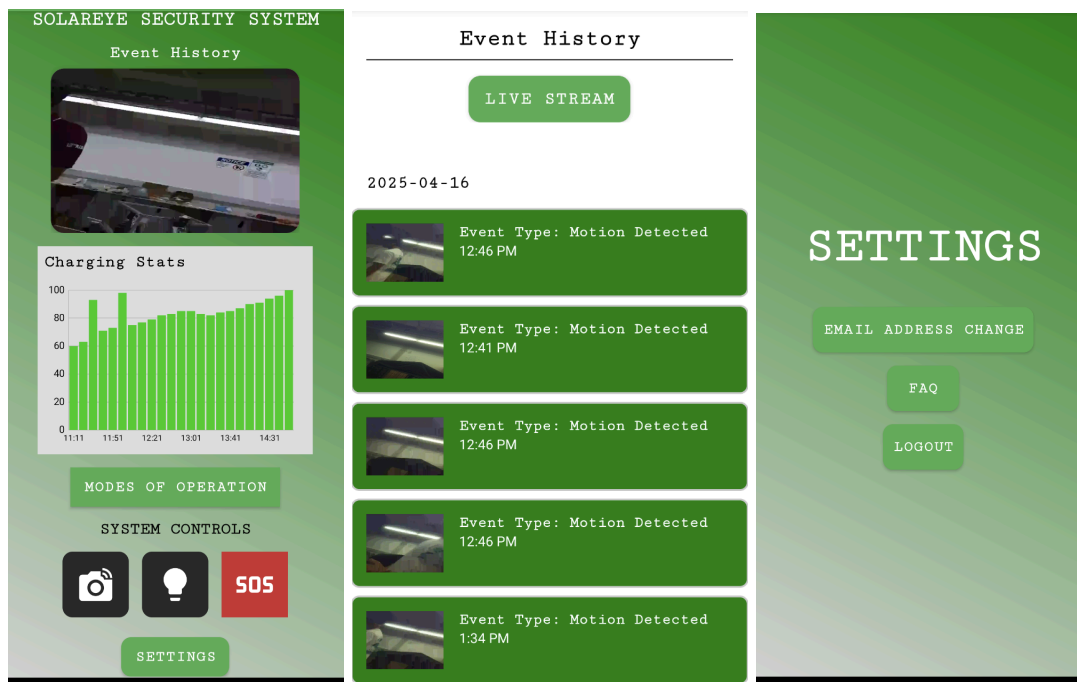


Figure 26: Application Home page with Charging Stats/Modes of Operation/System Controls, Event History page with Live Stream option, Settings Page with FAQ (APP VERSION ECEN 404)

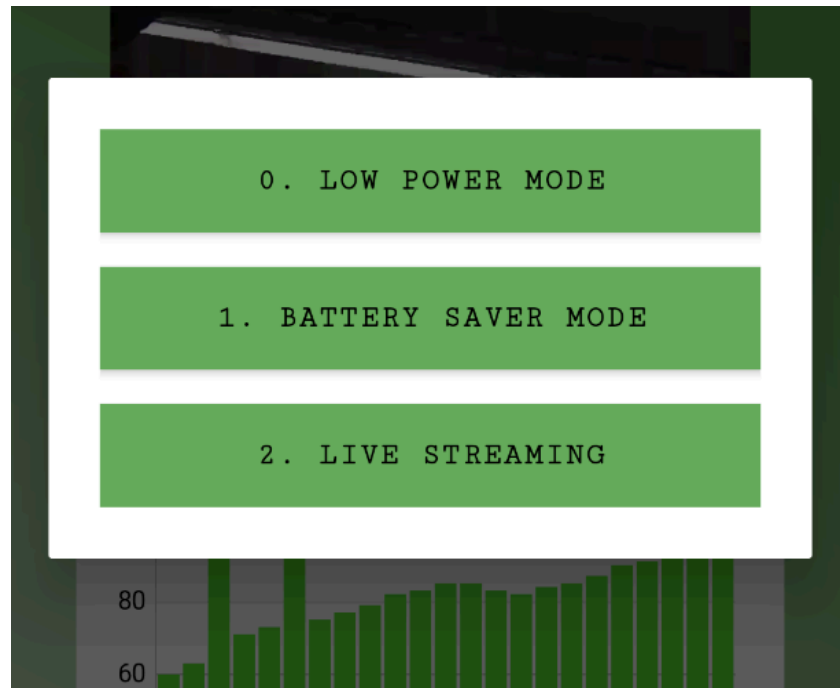
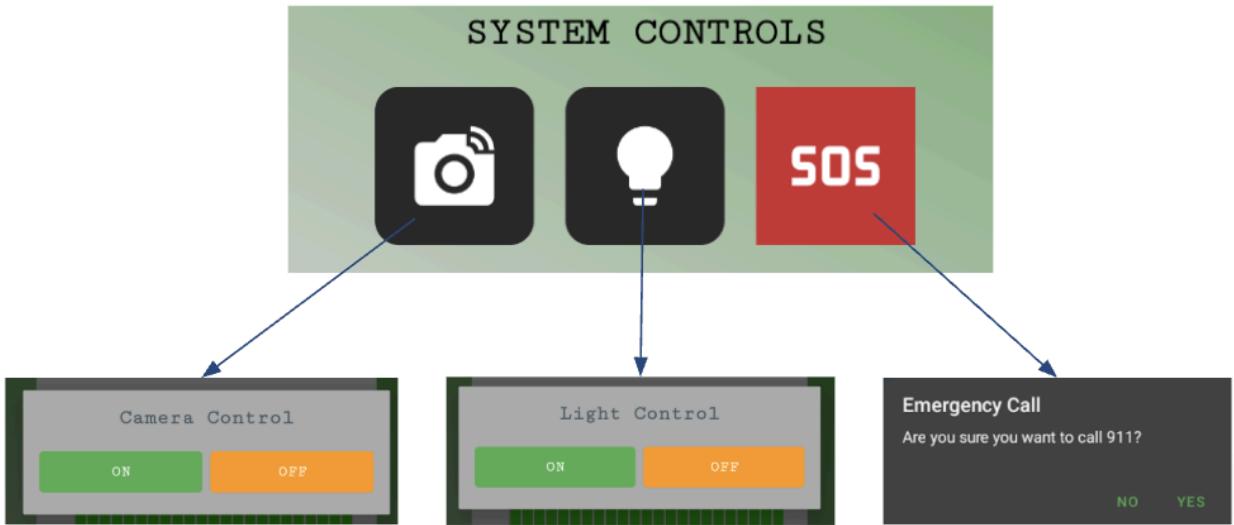


Figure 27: Modes of Operation for power consumption. (APP VERSION ECEN 404)



Figure 28: System Controls, Camera & LED are only operational, not functional (APP VERSION ECEN 403)



**Figure 29: System Controls, Camera & LED are only operational not functional
(APP VERSION ECEN 404)**

5.4 Android Application Conclusion

In conclusion, the SolarEye Security System app is fully operational and fully integrated with the microcontroller which is the only subsystem that the Android Application communicates with. This application enables the user to work on easy access and an interface-supported system, including a FAQ page for support. All precautions have been taken to ensure the security to only allow any authorized persons to create their account access for the system. Making use of Firebase APIs, everything remains encrypted at both mobile as well as the database interaction.

SolarEye Security System

Alex Allahar

Victor Igbokwe

Jackson Kitchel

Zayan Prasla

SYSTEMS REPORT

1. Complete System Overview

The SolarEye Security System is a comprehensive, solar-powered security system that aims to protect playgrounds and parks without the requirement of traditional power sources. The system combines solar energy harvesting, power management, edge processing, camera and sensor management, and a mobile application to provide an end-to-end and reliable surveillance platform.

A 100W solar panel charges a 12V battery with an MPPT charging circuit, giving constant, renewable power for 24/7 operation. Power is supplied through a bespoke buck converter and relay module that safely sends the correct voltages to the Raspberry Pi and other connected hardware.

At the core of the system, a Raspberry Pi 4B manages real-time action. It detects motion events with a PIR sensor, records video with an onboard camera, tracks battery and solar levels with an ADC, and controls fans and LED deterrence through GPIO pins. The system supports three user-configurable modes — Low Power Mode, Motion Detection Mode, and Livestream Mode — that enable users to opt for power preservation or real-time action depending on their requirements.

When movement is detected, the system captures video, sends an alert, and stores the event securely in Firebase. Users communicate with the system through the SolarEye Mobile App, where users can see events, adjust system settings, check battery status, and turn live streaming on when necessary.

A picture of the completed housing is here:

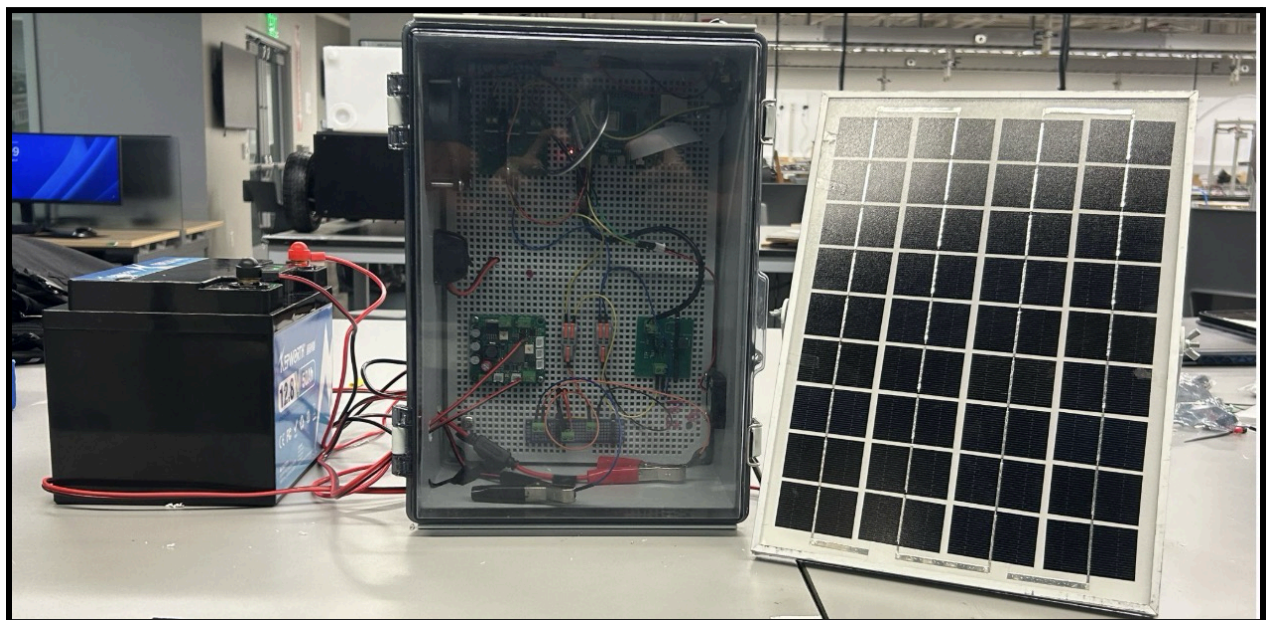


Figure 1: System Housing Overview

2. End-End System Scenarios

2.1 Loss of Solar Power and Battery Critically Low

On a cloudy day or overnight, when there's no sunlight available, the SolarEye System automatically switches to battery power. The solar panel stops charging, but the battery continues supplying energy, regulated by the power distribution circuits. The Raspberry Pi keeps monitoring the battery health, and if the charge drops below 20%, the system automatically enters Low Power Mode. In this mode, SolarEye still monitors for motion and flashes the LED deterrent when needed but disables video recording to save energy. The user gets a notification through the SolarEye App warning that the system is running on limited power. Once the sun rises again, the solar panel starts charging the battery, and the system automatically returns to full operation without needing user input.

2.2 Intruder Detected (Motion Detection and Event Capture)

If an intruder enters the monitored area, the PIR sensor detects motion and the Raspberry Pi activates the camera within 0.2 seconds. In near real-time, footage is recorded, uploaded to Firebase, and the user is notified through the SolarEye App. If LED deterrence is enabled, it flashes immediately to deter the intruder, ensuring fast reaction to security threats.

2.3 User Actively Viewing (Remote Livestream Monitoring)

When a user taps the Livestream button in the app, SolarEye starts streaming live video with minimal delay. An HTTP server is launched, and the video is made available almost immediately through a secure link. If solar power is available, it helps support the additional load; if not, the battery takes over, maintaining smooth, near real-time monitoring.